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CardioEnsemNet: An Optimized Ensemble Learning Framework for Accurate Coronary Artery Disease Prediction Across Multiple Datasets

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Abstract

One of the major causes of mortality around the world, poor diagnosis and the early stages of the disease being asymptomatic are the primary reasons for the occurrence of coronary artery disease (CAD). Therefore, early detection of CAD is essential to decrease the risk of death and improve clinical intervention. Machine learning (ML) methods have shown good performance for CAD prediction; however, existing methods face challenges such as limited cross-dataset generalizability, suboptimal feature selection, poor performance under class imbalance, and limited interpretability. To overcome these drawbacks, this study proposes a leakage-safe ensemble learning framework, namely CardioEnsemNet, for accurate and interpretable CAD prediction from heterogeneous clinical datasets. The framework is based on a combination of dataset-aware preprocessing, advanced feature engineering techniques, including polynomial and interaction features, Boruta feature selection, and SMOTE class balancing. To enhance predictive performance, the optimised stacking ensemble architecture is used, combining classifiers such as Random Forest, Support Vector Machine, K-Nearest Neighbours, XGBoost, and a meta-learner logistic regression. Furthermore, SHAP and LIME are added to provide interpretability of model predictions at both global and local scales. Four benchmark datasets, UCI Cleveland Heart Disease, Statlog Heart, Z-Alizadeh Sani and Framingham Heart Study were used to evaluate the proposed framework. Experimental results show that predictive performance is consistently high, with accuracy values exceeding 97%, F1-scores exceeding 95%, and ROC-AUC values close to 0.99 across all datasets. The robustness and contribution of the proposed preprocessing and ensemble learning strategies are then validated and confirmed by using cross-dataset validation and ablation studies. The results indicate that the CardioEnsemNet is a reliable, interpretable model for predicting CAD across diverse clinical data conditions.

Keywords Coronary Artery Disease · Ensemble Learning · Machine Learning · Explainable AI · Clinical Decision Support



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1 Introduction

CVD in general and CAD in particular cause around 18 million deaths annually, keeping CVD the leading cause of death worldwide. CAD — When coronary arteries become narrower, or blocked, by plaque, CAD usually progresses without causing any signs or symptoms until it has reached a critical stage, in which a heart attack, in most cases, is a deadly risk. Objectives: To determine the factors associated with a delayed (≥24 h post-admission) ACG and to characterise patients with CAD diagnosed>24 h after admission. Background: ML and AI are being explored as potential adjuncts for automatic CAD detection using clinical and physiological features. However, due to limited generalizability, class imbalance, and poor interpretability, they are often not used in the clinic [1, 2].

Literature on heart disease prediction using various ML models, such as DTs, SVMs, and ensemble methods [1, 2], but most of the work focuses on a single dataset and does not validate on multiple datasets. And more importantly, prior works frequently overlook these essential steps: (1) appropriate features from high-level groups, (2) class balance, and (3) psychiatric interpretability of prediction, which are all necessary for building clinically implementable models. However, the limitations of the existing literature further emphasise the importance of a transparent, generalizable framework that can leverage ensemble methods to improve CAD detection performance. To this end, we proposed a novel ensemble learning framework, CardioEnsemNet, that provides an optimal trade-off between prognostic performance and the interpretability of CAD predictions across multiple datasets. The study primarily targets (i) predicting a rigid ensemble framework for CAD detection, (ii) feature engineering and class imbalance considerations, and (iii) making the explanation using SHAP and LIME. Our contributions: (i) validation of the ooD ensemble both on a single dataset and across datasets, (ii) optimisation of meta-learners, (iii) combined integration for global and local interpretability.

The major contribution of this work is the proposal of an ensemble learning model that is systematically integrated and leakage-safe for predicting coronary artery disease across heterogeneous datasets. The study does not aim to develop a fundamentally novel learning algorithm used in isolation. Instead, it focuses on defining and integrating dataset-aware preprocessing, advanced feature engineering, class imbalance handling, optimised ensemble learning, explainability analysis, and cross-dataset validation into a consistent and reproducible CAD prediction pipeline. The framework aims to enhance robustness, interpretability, and generalisation capabilities across a wide range of clinical data scenarios.

The paper is structured as follows: Sect. 2 provides a comprehensive review of the relevant literature and highlights key gaps. The proposed CardioEnsemNet framework is outlined in Sect. 3, including datasets, preprocessing, feature engineering, and ensemble strategy. We present the experimental setup, dataset-wise evaluations, comparative analysis with base learners, cross-dataset validations, ablation studies, and explainability analysis in Sect. 4. Section 5 discusses the importance of the results and outlines the study's limitations. Acknowledgement Sect. 6 was the summary and discussion portion of the paper.

2 Related Work

In recent years, ML has enabled the development of various intelligent systems for predicting coronary artery disease (CAD). We categorise previous studies by individual classifiers, features, class imbalance, ensemble modelling, and generalizability across datasets, and review them in this section. This work can serve as a building block towards more practical, interpretable decision-support systems for predicting CAD that are accurate and robust to new, real-world datasets.

2.1 Machine Learning Approaches for Coronary Artery Disease Prediction

In many studies, classical ML models such as logistic regression, DTs, SVMs, and KNNs are used for early detection of CAD. Although these models are based on clinical and demographic features and have been shown to provide good baseline accuracy, their generalizability is often limited by their lack of adaptability to different data distributions. Ullah et al. An architecture based on optimal feature selection for CVD detection, employing machine learning, is scalable and 100% accurate but needs validation [1]. Khan et al. Deep learning models for CVD prediction: EnsCVDD-Net and BlCVDD-Net [2] achieved perfect prediction accuracy. Future efforts should mainly focus on appropriately handling unbalanced data and further validation. Singh et al. Our previous work in [3] presented a new ML-based cardiac disease detection approach and compared model accuracy and overfitting. Further studies should explore other methods to improve performance.

Anwer et al. Our work builds upon prior observational research [4] on ML approaches to enhance CTA for the detection of vascular stenosis, which, despite high accuracy, noted limitations. These limitations warrant expanding the patient population in future research. Hu et al. Gas-sensing approaches using machine learning have been reported as a non-invasive diagnostic method for cardiac heart disease [6]. Machine learning identified different types of breast molecules with 69.2% accuracy using light-activated technology. Baghdadi et al. To predict and manage early CVD [8], proposed a CatBoost model with high accuracy and F1-score. Sapra et al. They proposed using Gaussian noise to prevent overfitting [11] and achieved 96.2% accuracy in CAD diagnosis using deep neural networks. Bhan et al. ResNet, Vanilla-CNN, and Fourier-CNN were developed and compared, with successful segmentation of RV images for CVD diagnosis [14]. Chang et al. Please cite as: Gaurav Mehta & Jerusha Fernandez (2021) Development of a novel ML-based AI system to detect heart disease using Python: A discussion of accuracy, limitations, and need for further study, Health Information Science, 9, used a machine learning-based AI system in their paper for heart disease detec-tion, but the imple-mentation is also in the Python language and it has been discussed its accuracy, limitations, and need for further study [16]. Abdellatif et al. Many researchers have worked to improve CVD prediction using ML, SMOTE, and hyperparameter optimisation [18]. The result showed a very high level of accuracy, but the author also stated that further optimisation studies are still needed.

Using SMOTE and tuned ML classifiers, Dixit and Kala [21] achieved high accuracy with potential for improved early detection of CVD. Asadi et al. Found that MOPSO and Random Forest hybrids improved the accuracy of heart disease prediction by optimising the DT variety and training sets [29]. Wong et al. Wang et al. [31] reviewed the state of the art of DDN in cardiovascular diagnosis, summarising the challenges and directions to improve its clinical applicability. Li et al. This study [33] used an FCMIM-SVM system with complex feature selection methods to improve the detection speed and accuracy of cardiac disease. Stout et al. Virtual physiologic assessments that are more accurate and efficient compared to conventional evaluations can be used to direct virtual coronary interventions [34], Reddy et al. An enhanced feature extraction and precision approach based on an AGA and fuzzy logic (AGAFL) for predicting cardiac diseases earlier [35].

2.2 Feature Engineering and Selection Techniques in Cardiovascular Diagnostics

Feature Engineering for ML: Principles and Techniques for Data Scientists. Indeed, practical feature engineering appears to be an important step in improving model precision. Domain-driven feature selection, hybrid filters, embedded methods such as LASSO and tree-based selectors, PCA transformations, and the like, all result in smaller, more interpretable, and often faster feature spaces. Mahmood et al. The objective of this study was to create, compare, and evaluate the performance of various ML models for CAD prediction. Specifically, they found that a CAD prediction model based on an ensemble of Boosted DTs was the most successful, finalising plans for improved early detection and treatment of CAD [7]. Guo et al. A ML model introduced by [9] utilised echocardiography and indicated high sensitivity but low specificity in detecting CAD, which explains the emphasis on false positives. Uddin et al. When several ML algorithms were compared for predicting CD, the DT algorithm was found to be the best, with an accuracy of 99.16% [10].

Prabhu et al. In [13], quantum machine learning for multi-class CVD classification was proposed, demonstrating better performance with QNN and QSVC models. Sun et al. [15] conducted a review specifically focused on AI applications in cardiovascular disorders, summarising current applications, limitations, and directions for future research. Ahsan et al. Torres et al. [17] explored the challenges of using unbalanced data for cardiac disease prediction and machine learning, emphasising the problems this phenomenon poses and the need for future research. Qiao et al. The superiority of FFRCT over CCTA in reducing MACE and unnecessary referrals has been demonstrated in patients with intermediate coronary stenosis [20]. Ahmad et al. Random Forest and Decision Tree classifiers were presented by [23] as the most accurate, although they aimed to move to the next level of automatic heart disease prediction methods. The recent developments in heart disease prediction methods are compared. Abdollahi et al. PDQ shows that an ensemble classification model based on feature selection & evolutionary algorithms [24] achieves accuracy up to 97.57%. It also suggests directions for future research. Velusamy et al. CAD diagnosis through ensemble methods, combining high accuracy and reliability [27]. The SVM, RF, and KNN. Ghiasi et al. The CART algorithm employed in [30] to diagnose CAD proved highly accurate and robust, though it still has potential for further growth and adaptation. The MSSO-ANFIS model by Khan and Algarni [32], using more efficient optimisation methods, has achieved 99.45% accuracy in predicting cardiac disease.

Ahmed et al. developed a real-time prediction system for HD in [36] using Apache tools and comparing machine learning algorithms, achieving 94.9% accuracy with Random Forest. Rizzini et al. [38] However, there are anatomical and functional measurements that differ in lesion length, and we recently evaluated this in 38 [(see below)] and concluded that, due to these differences, they can affect PCI results. Future research should focus on integrating these assessments to optimise clinical decision-making. Lih et al. Using deep learning [41], proposed a 16-layer LSTM model with 98.5% accuracy in classifying ECG data for CAD, MI, and CHF. Further verification is required. Boonkaew et al. developed a prototype paper-based electrochemical system for ultrasensitive and multiplexed detection of cardiovascular biomarkers [45]. Limitations include potential scalability and field application issues.

2.3 Addressing Class Imbalance in Medical Datasets

For example, the healthy-hungry cohort has an advantage over two medical data sets at the CAD level, since it becomes increasingly common to have more healthy data than diseased data. They continue to assess whether solutions such as SMOTE, ADASYN, Tomek Links, and cost-sensitive classifiers can help balance representation of the minority class, thereby improving recall and overall precision-recall balance. Veeranjaneyulu et al. [12] predicted heart disease using MLP-EBMDA, achieving 87% accuracy, and compared this with an entropy-based random forest. Alilou et al. have, however, recently proposed a machine-learning solution to noninvasively identify atherosclerotic CAAs [19], which requires further incorporation and validation for clinical use. Javaid et al. Reichert et al. [22] focused on ML- based cardiovascular risk assessment [23] and reported the advantages and disadvantages of ML-driven risk stratification with respect to equity, safety, and privacy.

Ahmad et al. The best-performing heart disease prediction model, as identified by [25], was Extreme Gradient Boosting with GridSearchCV. It also provides recommendations for improving the model in the future. Shorewala et al. Multilevel Stacking with Feature Selection: A New Approach to Predicting Coronary Heart Disease" by [28] reviewed different ML techniques applied to CAD and announced that stacking models were the best-performing models, with guidelines for further enhancing performance. Kania et al. A non-invasive approach combining an algorithm to accelerate the evaluation of the diagnostic efficacy of a non-invasive method for detecting ischemic heart disease using fewer than the standard 12 leads in ECG [37]. Benhar et al. Markopoulos et al. [42] reviewed the various data preprocessing techniques, classification paradigms used for classifications of cardiac disease and highlighted the superior performance, but the challenges in clinical adoption. An effective IoT-based heart failure detection system using neutrosophic multicriteria decision-making was proposed by Son and Long [43], which alleviates confusion through a non-intuitive mechanism, thereby improving diagnostic accuracy. Su

et al. In another study [46, 47], similar to this one, the cardiac rehab was run by a nurse using eHealth, and the authors reported improvements in quality of life and health-related behaviours. Future research should address both the long-term effectiveness and the scalability of such programs. Serhani et al. As a result [48], achieved higher accuracy and offered patient-specific clinical recommendations for ECG-based arrhythmia typing using advanced CNNs. There is a need for additional investigations to improve validation in clinical practice.

Kazemian et al. In a 2022 literature review [50], examined the role of the gut microbiota in CAD, identified causal associations, and proposed multidisciplinary approaches to develop new therapeutic alternatives. Li et al. Fifty-four articles focused on the therapeutic benefits and disadvantages of curcumin and its cardiovascular effects and molecular targets. It also evaluates both preclinical and clinical research. Kramer et al. New Sections on Imaging with Devices [57] and an updated SCMR protocol will be updated periodically to reflect developments in new technologies. Khera et al. To counter COVID-19-induced disruptions in CVD treatment [60], proposed telemedicine, continuity of medication and flexible cardiac rehabilitation. Alizadehsani et al. [61](Invasive Procedures- Although not a standard operating procedure, coronary angiography remains an invasive procedure. Correct and reliable use of AI for CAD diagnosis brings two major benefits: lower cost and better patient care.

2.4 Ensemble and Hybrid Models for Cardiovascular Disease Detection

Several studies have also used ensemble learning strategies to improve predictive performance, such as bagging, boosting, and stacking. Hybrid architectures, which can include combinations of ML and DL components or multiple ML learners, are becoming common for producing a strong, interpretable computer-aided diagnosis (CAD) classification framework. Naeem et al. Lastly [5], presented a method based on an ANN trained on sensor data for person recognition, achieving over 85% accuracy but requiring ongoing improvement. Khiarak et al. An innovative competitive approach has been used by [26] to enhance the selection of heart disease-related characteristics that would harmonise improvements in accuracy while using a small number of features. Dhiman et al. A multi-objective deep learning model for fast and efficient COVID-19 identification from X-rays based on CNNs and a J48 decision tree [39]. Future research should examine the applicability and further development. Houssein et al. MRFO-SVM: A Novel Arrhythmias Classifier in ECG Signals Using Manta Ray Optimisation and Support Vector Machines [40] proposed MRFO-SVM, an ECG arrhythmias classifier based on Manta Ray Optimisation with Support Vector Machines (SVM) and achieved 98.26% classification accuracy. Future work should replicate it using multiple datasets.

Tuli et al. HealthFog is a low-latency framework for practical HD prediction that combines edge computing and ensemble deep learning [44]. One such limitation is balancing speed and accuracy. Alkhodari et al. Specifically, through deep learning on PCG data, it could identify valvular heart disorders with high accuracy [47]. Further research should test the durability and feasibility in a real-world clinical setting. Liu et al. DL algorithms are well recognised for achieving high accuracy in ECG diagnosis [49]; however, they also have limitations, and the authors therefore shared some directions for future work in deep learning-based ECG detection. Ahn et al. The CardioNet database, developed by Soares et al. [51], integrated unstructured EHR data in the CVD domain, facilitating the development of AI models. Limitations indicate the need for continuous progress in processing existing and future data to enable multicenter research with an improved, extended database. Soppert et al. Fat, lipoproteins, cardiovascular disease, and new and old treatments to control, prevent or counteract the dangers of fat [52]. Further research is needed to determine the best way to translate these problems into practical treatment approaches. Dong et al. After reviewing the evidence [53], explored the possibility that intermittent fasting might be beneficial for heart health and outlined guidelines for future research to confirm their hypothesis. Li et al. [54] highlighted the therapeutic potential of curcumin in cardiovascular disease management due to its antioxidant and anti-inflammatory properties. Dixit et al. Post-Acute COVID-19 Syndrome Heart Symptoms: Assessment Tools and Gaps in Knowledge (PPOS) [55].

Koushki et al. One recent study [56] conducted a comprehensive examination of how statins affect TLR and inflammasome pathways, which collectively contribute to atherosclerosis. It showed that statins' influence on

these pathways aligns with models of inflammation vs. disease progression. Real-time patient monitoring for diabetes improves diagnosis with lower latency via an energy-efficient cloud-edge system proposed by Abirami and Chitra [58]. Guo et al. The mAFA-II app is an example of dynamic HAS-BLED risk monitoring that reduced bleeding episodes and increased OAC use in patients with atrial fibrillation [59]. Li et al. proposed a new FFF using PCG signals and deep learning in [62], achieving 90.43% accuracy for CAD identification, which is higher than that of previous methods. Shahid et al. Specifically [65], showed that the PSO-ELM model with a subset of features selected via feature selection outperformed the standard ELM for CAD diagnosis, paving the way for further applications.

Ghasemi et al. With 303 patient samples and refined feature selection, the model achieves 99.67% accuracy in CAD diagnosis [68], but highlights the need for further dimensionality reduction. Glowacki et al. [70] broadly stated the promise of machine learning for screening CAD and suggested that noncontrast CT for Ao calcification reduces costs for both patients and the diagnosis. Although Lee et al. Despite concerns about data quality and consistency [72], AI-based CACS systems have the potential to enhance the utility of coronary artery calcium scoring.

2.5 Cross-Dataset Evaluation and Generalizability in Medical AI Models

The majority of studies do not explicitly test CAD models across multiple datasets, which is a crucial step towards achieving clinical trust and real-world applicability. Different strategies, such as cross-validation, domain adaptation, transfer learning, and meta-learning, are increasingly used to improve generalizability. Butun et al. They thought that, with >99% accuracy in CAD recognition achieved using the 1D-CADCapsNet model on short-duration ECG segments, timely and accurate diagnostics would be anticipated [63]. AlSafi and Ocan [64] demonstrated that their CHD detection model achieved an accuracy of over 90%, surpassing prior approaches, by using improved ML algorithms and feature selection. Saharan et al. They used cytokine biomarkers and ML methods (such as k-NN and Random Forest)—i.e., a non-invasive, accurate, and inexpensive diagnosis of coronary artery disease CAD [66]. These techniques make it more effective than a traditional one. Das et al. Using multiple feature selection algorithms and 21 features [67], employed an advanced approach and achieved an accuracy of 92.31% in CAD classification.

A hybrid PSO-EmNN model for CAD diagnosis was developed by Shahid and Singh [69] with a maximum accuracy of 88.34%. Other developments and more algorithms were proposed. Nous et al. While the impact of off-site CT-FFR on clinical decision-making is still to be defined [71], has shown that on-site CT-FFR enhances the clinical diagnostic yield for obstructive CAD. Xiao et al. The improved 3D U-Net for coronary artery segmentation addresses the challenges of image detail and data expansion identified by [73] and achieves optimal accuracy with centerline data. Zhang et al. To improve data quality and generalizability [74], developed a two-step stacking model for early CHD detection based on 2D-STE and clinical characteristics. Jahmunah et al. proposed a GaborCNN model that showed improved results than a CNN in classifying normal, CAD, MI, and CHF from ECG data [75]. Its precision exceeds 98.5%. Kawasaki et al. Though limited by a small patient cohort, integration of CCTA with CT-FFR and machine learning has been shown to improve CAD diagnosis [76]. Nilashi et al. Incremental learning reduces time consumption and increases precision in heart disease prediction; however, larger datasets are required [77]. Umar Khan et al. The study used SVMs and ECG data to predict CAD (from 27,552 records from the UCI ML repository), yielding an accuracy of 95.5% [78], with promise for future deep learning breakthroughs. Pathak et al. Shortly after, Wan et al. [79] developed a powerful PCG-gradual CAD identification system with a planar-like sensitivity of 76.5%; however, this system should be evaluated under different noise conditions. Wang et al. The two-level machine learning model used by [80] achieved 95.43% accuracy on CHD when the data were split into training and test sets; achieving proper training time and model tuning remains a challenge.

Another recent study on the importance of combining feature selection, classifier optimisation, and ensemble learning for cardiovascular prediction used optimisation methods. Kumar and Rekha [81] have used an improved

Hawks optimiser-based learning method for cardiovascular disease prediction, demonstrating that optimising feature selection and classification can enhance classifier accuracy. Arunachalam and Rekha [82] developed a machine-learning-based cardiovascular disease prediction model using ensemble classifiers and found that combining various learners effectively boosted diagnostic performance.

The existing literature on CAD prediction modelling shows generalisation inabilities, class imbalance, and interpretability issues, which are treated as weaknesses in the study. CardioEnsemNet is explicitly designed to address these gaps by incorporating state-of-the-art feature engineering, ensemble learning, and explainability. Past literature provides insights that not only motivate but also justify the novelty of the proposed contributions to the framework.

3 Materials and Methods

Here, we will outline the overall methodology for building the CardioEnsemNet framework in detail. It elaborates on the datasets used, preprocessing and feature engineering techniques, and various approaches to addressing class imbalance. It also details the ensemble architecture, model training, and explainability methods used to maintain high prediction accuracy, robustness, and clinical relevance across datasets of coronary artery disease that are similar yet distinct.

3.1 Overview of CardioEnsemNet

The proposed CardioEnsemNet framework is a multi-stage pipeline that enhances prediction robustness, optimises feature selection, handles class imbalance, and improves interpretability across heterogeneous coronary artery disease datasets. The framework comprises six major stages: (i) dataset-aware preprocessing, (ii) advanced feature engineering, (iii) addressing class imbalance through SMOTE, (iv) optimised ensemble learning, (v) explainability analysis using SHAP and LIME, and (vi) cross-dataset validation using the meta-ensemble aggregation. The stages aim to address various issues associated with current CAD prediction systems, such as limited feature representation, data imbalance, poor generalisation, and lack of interpretability.

To address the inherent class imbalance in medical datasets and ensure balanced learning, the Fake Minority Over-Sampling Technique (SMOTE) [10] was employed. Then an ensemble with optimised base classifiers is built on the preprocessed and balanced dataset, comprising RF, SVM, KNN, and XGBoost. We are using GridSearchCV to tune the hyperparameters of each base learner, ensuring smooth tuning.

The whole workflow of the proposed CardioEnsemNet framework is presented in Fig. 1. The pipeline starts with data acquisition and preprocessing, followed by feature engineering and optimisation. Then the processed data are balanced with SMOTE and sent to several optimised base classifiers. The outputs are combined using stacking and weighted soft voting, and interpretability is provided at both global and local levels via SHAP and LIME. Last but not least, cross-dataset validation and meta-ensemble learning are conducted to assess robustness across diverse clinical datasets with varying characteristics.

The proposed framework first performs preprocessing steps on the datasets, including imputation of missing values, encoding of categorical values, feature scaling, and stratified splitting. To boost feature relevance and eliminate redundancy, advanced feature engineering is employed, including polynomial feature generation and Boruta feature selection. Due to class imbalance, SMOTE is used only on the training data. The best-performing features are then shared with four base classifiers: Random Forest, SVM, KNN, and XGBoost. The outputs from these are combined using weighted soft voting and a logistic regression meta-learner to arrive at the final prediction. SHAP analysis and LIME are used for explainability and cross-dataset validation to evaluate generalisation capability.

The novelty of CardioEnsemNet lies mainly in the structured integration and optimisation of multiple complementary machine learning components within a single framework for heterogeneous CAD prediction. The study

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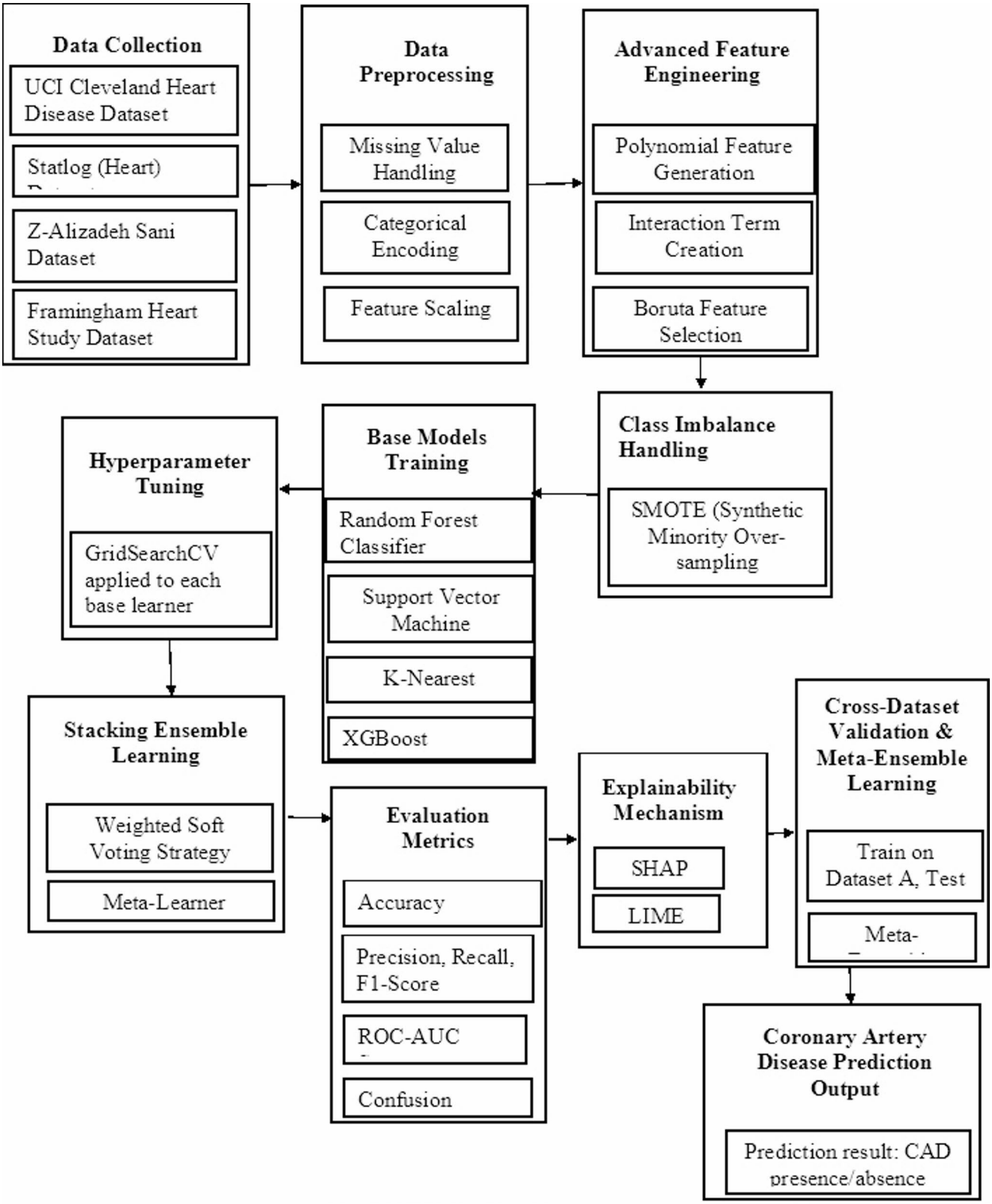


Fig. 1 Detailed Architecture of CardioEnsemNet Framework for Coronary Artery Disease Prediction, Highlighting Each Stage and Sub-Stage

focuses on strong empirical engineering, reproducible preprocessing, and cross-dataset evaluation, explainable ensemble learning, and proposes nothing revolutionary for the base learning algorithm.

3.2 Datasets Usage Strategy

In this paper, we evaluate the power and generalizability of the proposed CardioEnsemNet for predicting coronary artery disease (CAD) using four publicly available benchmark datasets. Such datasets contain multiple demographic and clinical variables needed for replicability in other population cohorts. Characteristics of the individual datasets are summarised in Table 1.

3.2.1 UCI Cleveland Heart Disease Dataset

The UCI Cleveland HD dataset is among the most popular for CAD prediction research. It includes 303 patient records, each described by a sum of 13 clinical attributes (age, sex, resting BP, cholesterol, FBS, maximum heart rate, exercise-induced angina, etc.). The target variable is also binary: whether or not heart disease is present.

3.2.2 Statlog (Heart) Dataset

The Statlog (Heart) dataset contains the same 13 clinical attributes, has 270 instances, and is based on the Cleveland dataset. It has a slightly different distribution and is frequently used as a benchmark for ML algorithms. As in the Cleveland dataset, the target variable follows a binary classification scheme.

3.2.3 Z-Alizadeh Sani Dataset

The Z-Alizadeh Sani dataset is newer and more comprehensive, comprising 303 records and 54 descriptive attributes, including demographic, clinical laboratory, and ECG features [1]. It is widely used in CAD diagnosis and has a larger feature set than other datasets. Finally, the target variable is binary, indicating the patient's CAD diagnosis status (yes/no).

3.2.4 Framingham Heart Study Dataset

Framingham Heart Study dataset. This dataset comprises 4,240 instances and 15 attributes, providing demographic information for patients and clinical parameters such as BP, BMI, smoking status, diabetes, and others. The dataset is used for assessing LTCVD risk. The target variable, or dependent variable, is binary, indicating whether or not the patient develops coronary heart disease within 10 years.

Table 1 Description of the 4 benchmark datasets used for experimental analysis in this study, with varying levels of sizes and feature dimensions, with the targeted variable of the dataset about heart disease or coronary risk prediction. Notations: Table 2 includes crucial notations we will adopt in the methodology, such as variables and functions, that help clarify how an algorithm or model works.

Sample size, dimensionality, type of clinical variables and interpretation of the target differ across the four datasets used in this study. There are 13 clinical features in Cleveland, 54 demographic, clinical, laboratory,

Table 1 Summary of Datasets Used

Dataset Name	Number of Instances	Number of Features	Target Variable
UCI Cleveland Heart Disease	303	13	Presence of Heart Disease (Binary)
Statlog (Heart)	270	13	Presence of Heart Disease (Binary)
Z-Alizadeh Sani Dataset	303	54	Coronary Artery Disease Diagnosis
Framingham Heart Study	4240	15	10-Year Coronary Heart Disease Risk

Table 2 Summary of Notations Used in CardioEnsemNet Framework

Notation	Description
X	Original feature vector of an instance.
X_{poly}	Polynomially expanded feature vector including squared terms and interaction features.
X_i	Value of the ithi^{th} instance's feature.
$X_i^{imputed}$	Imputed value for the ithi^{th} feature after median-based missing value handling.
X_i^{scaled}	Scaled feature value after applying StandardScaler normalization.
μ	Mean of a feature used in normalization.
σ	Standard deviation of a feature used in normalization.
C	Categorical feature with kk distinct categories.
B_{ij}	Binary value in One-Hot Encoding matrix for instance ii and category jj.
Z_{orig}	Feature importance score of an original feature in Boruta selection.
Z_{shadow}	Maximum feature importance score of shadow features in Boruta selection.
D_{train}	Training dataset subset.
D_{test}	Testing dataset subset.
h_i	Base learner model (Random Forest, SVM, KNN, XGBoost) in the ensemble.
$P_i\,(c)$	Predicted probability for class cc from the ithi^{th} base learner.
w_i	Weight assigned to the ithi^{th} base learner in weighted soft voting.
$P_{final}\,(c)$	Final predicted probability for class cc after ensemble voting.
$\varnothing_i$	Shapley value contribution of the ithi^{th} feature in SHAP explanation.
f(x)	Model output for instance xx in SHAP explanation.
x_{new}	Synthetic instance generated by SMOTE.
x_i	Original minority class instance in SMOTE.
x_{nn}	Nearest neighbor instance in SMOTE.
δ	Random number between 0 and 1 used in SMOTE synthetic sample generation.
$\mathrm{P_{meta}}\,(\mathrm{c})$	Final predicted probability after meta-ensemble aggregation across datasets.
N	Total number of datasets used in meta-ensemble learning.

and ECG-related features in Z-Alizadeh Sani, and 15 features in Framingham that are used to predict 10-year coronary heart disease risk. Thus, no preprocessing was performed as a blind uniform transformation. Rather, a dataset-aware preprocessing approach was employed, in which each dataset was preprocessed according to its feature structure and target distribution.

3.3 Data Preprocessing

The four benchmark datasets feature varying dimensions of their features, clinical variables, patterns of missing values, categorical variables, and definitions of their targets, which makes the data preprocessing difficult. Because of the variety of dimensions of features, clinical variables, missing values, categorical variables, and definitions of target, the data preprocessing is difficult. The data sets were treated independently prior to training the models. Imputing (median), one-hot encoding and standard scaling were applied to numerical missing data, numerical attributes and categorical attributes, respectively. Imputation statistics, encoding mappings, scaling parameters, features selection and oversampling methods were learned from training data and applied to the validation or test data to prevent information leakage.

First, median imputation is used to handle missing values, which are often seen in medical datasets based on various clinical conditions. Any missing numerical feature is replaced by its median value; in this way, we minimize the influence of outliers. This can be expressed mathematically as in Eq. 1.

$$\mathrm{X_i^{inputed}} = \begin{cases} \mathrm{X_i} \\ Median\ (\mathrm{X_{feature}}) \end{cases} \quad \text{if } \mathrm{X_i} \text{ is not missing} \tag{1}$$

Where $\mathrm{X_i}$ denote the value of the $\mathrm{i^{th}}$ Instance.

We will perform one-hot encoding on categorical variables (helicopter area and pool area) to convert them to a 0/1 and machine learning-friendly format. Assume CC is a categorical feature with kk classes. CC is then converted into a binary matrix BB of shape kk using One-Hot Encoding as in Eq. 2.

$$B_{ij} = \begin{cases} 1 & if\ instance\ i\ belongs\ to\ category\ j \\ 0 & otherwise \end{cases} \tag{2}$$

This transformation tricks the model into assuming no ordering relationship between categories in categorical data.

We used standard normalization for feature scaling to ensure the numerical feature space has no bias due to differences in scale. For computing the mean and variance, the following formula is used to scale each feature to have a mean of zero and a variance of one, as per Eq. 3.

$$X_i^{scaled} = \frac{X_i - \mu}{\sigma} \tag{3}$$

Where μ is the mean and σ is the standard deviation of the corresponding feature.

After finishing the preprocessing step, each dataset underwent stratified splitting into training and testing subsets to maintain balance between classes. In particular, a 70:30 train-test split is used to ensure a balanced representation between positive and negative classes. More formally, we can express this as in Eq. 4.

$$D = D_{train} \cup D_{test} \text{ such that } |D_{train}| = 0.7\,|D|, |D_{test}| = 0.3\,|D| \tag{4}$$

In particular, a uniform preprocessing pipeline is applied to all datasets, including missing value imputation, categorical encoding, feature scaling, and stratified splitting, which guarantees data quality and consistency and contributes to the proposed CardioEnsemNet framework.

3.3.1 Dataset-Aware Preprocessing and Leakage Prevention

Because of the different natures of the data sets, no common preprocessing was done across all data sets, but rather preprocessing was done separately for each data set, without forcing an identical raw feature structure of all data sets. Within-dataset evaluation involved splitting each dataset into parts by stratified splitting to maintain the class distribution. Next, all data-dependent preprocessing operations were fitted only on the training data set. To ensure that the results are solely based on the training data, median values for imputation, category mappings for one-hot encoding, mean and standard deviation for scaling, Boruta feature selection and SMOTE-based oversampling were calculated only from the training data. The learned transformations were then used in the test data without retraining.

To ensure cross-dataset evaluation, a harmonized feature representation was used, when common clinical features were available. If the data sets did not contain overlapping attributes, the model was only evaluated after consistently applying the relevant preprocessing and feature transformation. This way, target-related information from the test data set was not used, and the risk of over-optimistic estimation of performance was minimized. SMOTE was only used for the training data and never validation or test data.

The feature dimensionality, the variable composition, and the definition of clinical attributes vary greatly between the benchmark datasets used in this study. There are 13 main clinical attributes in Cleveland, 54 demographic and laboratory-related attributes in Z-Alizadeh Sani, and 15 long-term cardiovascular risk indicators in Framingham. Thus, direct transfer of raw features from one dataset to another was not carried out.

Where possible, a consistent approach to feature representation was taken, utilizing clinically relevant and readily available cardiovascular metrics such as age, sex, blood pressure measures, measures related to cholesterol levels, heart rate metrics, diabetes metrics and clinical metrics related to exercise. The features that were not provided, and were not matched in, across datasets were not directly transferred and evaluated.

Moreover, the pre-processing and feature scaling, the polynomial feature generation and feature selection with Boruta algorithm were done separately for each dataset, utilizing only the respective training data. Consequently, the feature subsets optimized by Boruta were dataset-specific and were not expected to be the same in two different data sets. The cross-dataset experiments thus examined the robustness and transferability of the ensemble learning framework for a non-strict feature-level interoperability, but rather for heterogeneous cardiovascular prediction settings.

3.4 Advanced Feature Engineering

The proposed CardioEnsemNet framework utilizes an enhanced and sophisticated feature engineering approach to enhance the model's learning capabilities. It systematically increments the original feature space and determines the most relevant predictors. This essential procedure enables the examination of multi-linear relations and interaction effects between clinical variables that may influence the prediction of CAD.

In Fig. 2 "Polynomial feature generation" is first performed to add higher-order terms and interaction features to the feature set. In particular, polynomial features of degree 2 are created for each feature vector XX. We refer to this expansion of the feature space $X = [x_1, x_2, \ldots ., x_n]$ as a new space composed of all squared and pairwise interactions, expressed mathematically as in Eq. 5.

$$X_{poly} = [x_1, x_2, .., x_n, x_1^2, x_2^2, \ldots , x_n^2, x_1x_2, x_1x_3, \ldots , x_{n-1}x_n] \tag{5}$$

Polynomial and interaction terms are included, allowing for the capture of non-linear dependencies and interactions between features often present in medical data.

The other thing was that after expanding the feature space, we would then prune the feature space to remove useless or duplicate features, compress the dimensionality, and prevent overfitting. For this purpose, it uses the Boruta feature selection algorithm. Boruta is a comprehensive feature selection technique based on the feature importance scores from the Random Forest classifier. This method iteratively compares the original features and the randomly permuted shadow features' importance, keeping only those that demonstrate significantly more significant importance. More formally, let Z_{orig} be the importance score of an original feature and Z_{shadow} be the maximum importance score of shadow features; a feature is selected as in Eq. 6.

$$Z_{orig} > Z_{shadow} \quad \text{with statisical significance} \tag{6}$$

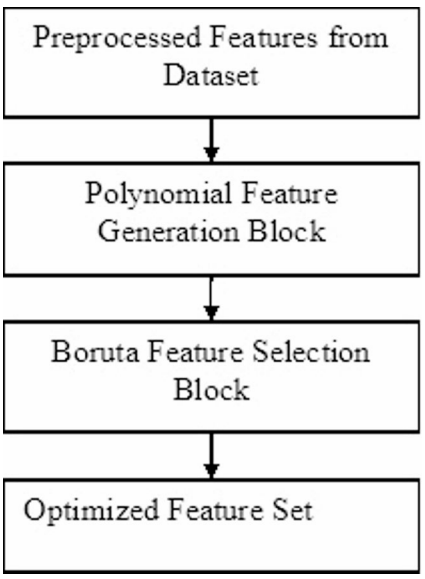


Fig. 2 The Advanced Feature Engineering Process Flow in CardioEnsemNet

So, the set of features used for learning ensemble classifiers was enriched and optimized using polynomial feature generation and Boruta-based feature selection, respectively, to enable throughput with computational efficiency. This fine-grained feature engineering is essential for boosting the predictive performance of CardioEnsemNet as a whole.

The feature engineering stage significantly improved the representation of the clinical attributes and created polynomial and interaction-based features that could capture non-linear relationships between the cardiovascular risk indicators. Then, the redundant and low importance features were eliminated using feature selection based on Boruta, leaving a small and informative subset of features which would be used for training the models. The preprocessing and feature optimization steps in total helped in obtaining consistent features, noise reduction, and learning stability of the ensemble classifiers for heterogeneous datasets.

The datasets contained different features and number of features, thus feature engineering and feature selection using Boruta were done independently for each of the considered datasets. This led to different sets of optimized features for each dataset, depending on the statistical significance and availability of the clinical parameters in the specific data set.

3.5 Handling Class Imbalance

Unbalanced classes in medical datasetLearning with such imbalanced classes is a common problem in medical datasets, especially the disease prediction datasets where we have many healthy (negative) cases and much fewer diseased (positive) cases. This imbalance can introduce bias in the machine learning models towards the majority, resulting in sub-optimal predictive performance for detecting minority class cases, which are typically the clinically relevant types.

To address this issue within the current CardioEnsemNet framework, the SMOTE is employed. SMOTE is another oversampling technique, but rather than duplicating samples from the minority class, new samples are generated in the feature space. This process works by selecting a minority class instance, determining its KNN, and synthesizing example instances between the selected instance and its neighbors along the line segments connecting them. A synthetic sample x_{new} is obtained as follows mathematically, as in Eq. 7.

$$x_{new} = x_i + \delta \times (x_{nn} - x_i) \tag{7}$$

where x_i is a minority class instance, one of its KNN, and δ is a random number between 0 and 1.

This method helps balance the data set without replicating instances, enabling the classifiers to clearly learn the decision boundary between the classes. After preprocessing and feature engineering, SMOTE is consistently applied to each of the four datasets so that the ensemble models trained on the balanced data can accurately detect coronary artery disease cases.

The benchmark datasets used in this study are relatively small, typically clinical tabular datasets and therefore cannot use traditional image augmentation methods like rotation, flipping or transformation. Hence, SMOTE is used here as a data-level augmentation method to synthetically augment minority class instances. This way it improves class representation, it decreases the model's bias towards the majority class and it enriches the learning ability of the ensemble classifiers without having to duplicate existing records.

By using SMOTE, the minority-class samples were better represented without adding more instances to them, which minimized classifiers' bias toward the majority classes. This resulted in better recall and equal predictive performance of the CAD in datasets of varying class distribution, especially in clinically relevant positive CAD cases.

3.6 Ensemble Learning Framework

The ensemble learning stage aims to fuse complementary features of various machine learning classifiers and minimize learning model bias and variance. The main idea of the CardioEnsemNet framework is to combine predictions from heterogeneous learners using stacking and weighted voting to build a more stable and robust predictive model across different datasets.

No metaheuristic optimization algorithm (like TAOA, TPSN) is used in the proposed CardioEnsemNet framework. Rather, hyperparameter tuning was performed for all base classifiers using GridSearchCV with cross-validation and stratification. Optimal parameters for each of the models (Random Forest, SVM, KNN and XGBoost) were determined using the validation performance to ensure that model tuning was stable and reproducible across datasets.

At the heart of the proposed CardioEnsemNet framework is an optimally tuned ensemble learning methodology that combines the strengths of diverse base classifiers to accurately predict coronary artery disease. The prediction power of various classifiers is combined to reduce variance, minimize bias, and thus enhance accuracy. In this research, we consider four diverse base classifiers, which include RF, SVM, KNN, and XGBoost. One by one, each base learner is tuned with GridSearchCV hyperparameter optimization for the best performance. The trained RF, SVM, KNN, and boost models are represented by h1, h2, h3, h4. The predictions of these base learners are then combined using a stacking ensemble pea approach with weighted soft voting. Soft voting means multiplying the probability of each base learner by its predefined weights and summing to form the final prediction. Formally, we refer to the final predicted probability $P_{final}(c)$ for class c as in Eq. 8.

$$P_{final}(c) = \sum {}_{i-1}^{4} w_i \times P_i(c)$$

where is the weight attached to the ith base learner, and $P_i(c)$ is the learner's predicted probability for class i^{th} given learner will have a different weight based on its performance on the validation set, voted by giving great weight to more accurate models.

To improve the ensemble even further, a logistic regression meta-learner is trained to learn from the predictions of the base learners. The discrimination power of the CardioEnsemNet framework is formulated based on an ensemble of a number of base classifiers trained independently and an optimal combination strategy, which benefits from the diversity of the base classifiers and the optimal strategy.

3.7 Proposed Algorithms

The CardioEnsemNet framework is operationalized through a sequence of five core algorithms, each addressing a critical stage in the coronary artery disease prediction pipeline. These algorithms encompass data preprocessing, feature engineering, class imbalance handling, ensemble model training, and cross-dataset validation. Together, they enable robust, scalable, and generalizable CAD prediction across multiple clinical datasets with improved interpretability.

The entire operational pathway of the CardioEnsemNet is depicted in five organized algorithms, that encompass the preprocessing, the feature engineering, the imbalance treatment, the ensemble learning, and the cross-dataset evaluation.

Algorithm: Leakage-Safe Data Preprocessing Procedure
Input: Raw dataset D
Output: Preprocessed training set $D_{train}^{processed}$, testing set $D_{test}^{processed}$

1. Split dataset D into stratified training set D_{train} and testing set D_{test} using a 70:30 ratio while preserving class distribution.
2. For each numerical feature in D_{train}:
   - Compute the median value from training data only.
   - Replace missing values in both D_{train} and D_{test} using the training median.
3. For each categorical feature in D_{train}:
   - Fit One-Hot Encoding using training categories only.
   - Apply the learned encoding transformation to both D_{train} and D_{test}.
4. Fit StandardScaler on numerical features of D_{train}:

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

where μ and σ are computed only from the training data.

5. Transform numerical features in both D_{train} and D_{test} using the learned scaling parameters.
6. Store preprocessing parameters separately for each dataset to accommodate heterogeneous feature structures and variable distributions.
7. Return leakage-safe preprocessed datasets $D_{train}^{processed}$ and $D_{test}^{processed}$.

Algorithm 1: Leakage-Safe Data Preprocessing Procedure

The dataset-aware learning pipeline with leakage safety is ensured by applying Algorithm 1 for the prediction of coronary artery disease.For the prediction of coronary artery disease, Algorithm 1 is applied to ensure a leakage safe and dataset aware learning pipeline. Given that the benchmark datasets have different feature dimensions, types of variables, and clinical distributions, preprocessing is done separately for each dataset. To avoid information leakage, stratified splitting is utilized prior to all the pre-processing tasks. Missing value imputation, categorical encoding and feature scaling are fitted only on the training data, and then applied to the test data with parameters learned from the training data. This design will retain the validity of model evaluation and compatibility with heterogeneous clinical data.

Algorithm: Advanced Feature Engineering Strategy
Input: Preprocessed feature set $X_{preprocessed}$
Output: Optimized feature set $X_{optimized}$

1. Apply polynomial feature expansion to $X_{preprocessed}$, generating new features up to degree 2:
   Include squared terms and pairwise interaction terms.
2. Initialize a Random Forest classifier to compute feature importance scores.
3. Apply Boruta feature selection:
   For each original feature, compare its importance score Z_{orgi} with the maximum importance score of shadow features Z_{shadow}.
   Retain original features where $Z_{orgi} > Z_{shadow}$. with statistical significance.
4. Remove irrelevant and redundant features based on Boruta selection.
5. Return the optimized feature set $X_{optimized}$.

Algorithm 2: Advanced Feature Engineering Strategy

Feature engineering to improve CAD models, as implemented in Algorithm 2. Initially, it creates a larger input feature set by applying polynomial transformations, which generate new features by multiplying existing features to create squared and interaction terms. This expansion contributes to the capture of non-linear relationships between clinical variables. Next, to prevent overfitting and decrease dimensionality, the Boruta feature selection method is employed based on the importance scores from a RFmodel. It retains a feature only if it has consistently greater importance than its randomly permuted counterparts, eliminating spurious attributes and retaining only the relevant ones. The result is a much more informative and lean feature representation that together preserves the predictive information while maintaining a relatively short length to meet the requirement of efficient learning.

Algorithm: Class Imbalance Handling Using SMOTE
Input: Feature set X, target labels y
Output: Balanced feature set $X_{balanced}$, labels $y_{balanced}$

1. Identify the minority class samples in y.
2. For each minority class instance x_i:
   Find KNNof x_i in the feature space.
   Randomly select one neighbor x_{nn}.
3. Generate synthetic samples:
   For each selected neighbor, create a synthetic sample using:

$$x_{new} = x_i + \delta \times (x_{nn} - x_i)$$

where δ is a random number in the range (0,1).

4. Add the synthetic samples to the original dataset.
5. Return the balanced feature set $X_{balanced}$ and corresponding labels $y_{balanced}$.

Algorithm 3: Class Imbalance Handling Using SMOTE

Coronary artery disease datasets typically have a few positive cases and many negative cases, resulting in a class imbalance problem, which will be addressed in Algorithm 3. To balance it out, the algorithm relies on the SMOTE, which does not simply duplicate some of the instances of the minority class; instead, it generates samples of the minority class in an artificial manner. It works by observing the feature space of minority class samples and generating synthetic instances that bridge the gap between these existing ones and their nearest neighbors. This leads to a balanced dataset where both classes are sufficiently represented, allowing classifiers to learn characteristics from both classes and mitigating the bias toward the majority class. This balanced dataset is used in the next steps of model training to ensure that the prediction performance is fair and realistic.

Algorithm: Optimized Ensemble Learning Framework (CardioEnsemNet)
Input: Optimized feature set $X_{optimized}$, target labels y
Output: Trained CardioEnsemNet model

1. Initialize four base classifiers:
   RF
   SVM
   KNN
   XGBoost
2. For each base classifier:
   Perform hyperparameter tuning using GridSearchCV to identify optimal parameters.
   Train the classifier on $X_{optimized}$.
3. Combine the outputs of all base classifiers using a weighted soft voting strategy:

$$P_{final}(c) = \sum_{i-1}^{4} w_i \times P_i(c)$$

where w_i is the weight assigned to the ith base learner and $P_i(c)$ is the predicted probability for class c.

4. Train a meta-learner (Logistic Regression) on the outputs of the base classifiers to further refine the final prediction.
5. Return the trained CardioEnsemNet model.

Algorithm 4: Optimized Ensemble Learning Framework (CardioEnsemNet)

Section 5 highlights the CardioEnsemNet framework, utilizing the learning architecture of CardioEnsemNet, which involves the general integration of different optimized classifiers with CardioEnsemNet for fine-tuning to achieve better and standardized coronary artery disease prediction (the core learning architecture of the Cardio-EnsemNet framework). First, we initiate four base learners: Random Forest, Support Vector Machine, K-Nearest Neighbors, and XGBoost. A grid search strategy is applied to each base classifier for hyperparameter tuning to find the optimal configurations. These are the tuned models, which are trained on the best set of features produced after the preprocessing and feature engineering steps have been completed. At the end of the training, the predictions made by all base models are summed using the weighted soft voting approach, wherein the contribution of each model is weighed based on its performance. A meta-learner (usually logistic regression) is trained on the outputs of the base classifiers to improve the decision boundary. The outcome is a powerful ensemble model that

can learn both linear and non-linear relationships, as well as the trade-off between ensemble diversity (the difference between models) and ensemble generalization (the mean of the models' predictions).

Algorithm: Cross-Dataset Validation and Meta-Ensemble Aggregation
Input: Four datasets $D_1, D_2, D_3. D_4$
Output: Final aggregated prediction results

1. For each datasetD_i where i=1to4:
   Train the CardioEnsemNet model on D_i.
   Evaluate the trained model on the remaining datasets D_j where j≠ i.
   Record evaluation metrics (Accuracy, ROC-AUC, F1-score, etc.) for each cross-dataset evaluation.
2. For each test instance in the evaluation datasets:
   Collect predicted probabilities $P_i(c)$ from the models trained on all datasets.
3. Aggregate the predictions using meta-ensemble strategy:

$$P_{meta}(c) = \frac{1}{N}\sum_{i-1}^{N} P_i(c)$$

where N=4 (total number of datasets).

4. Output the final aggregated prediction results and evaluation metrics.

Algorithm 5: Cross-Dataset Validation and Meta-Ensemble Aggregation

We utilize this in our method, described in Algorithm 5, to test the generalizability and stability of our CardioEnsemNet framework at several datasets. For the split ensemble, we first train the ensemble model separately on each data-set. Each of these models are then evaluated on the other datasets that the model was not trained with. We perform this evaluation on two datasets (not only on a single dataset, which merely tests if the proposed approach fits the characteristics of a specific dataset), which also represents a more realistic use case on the model's suitability. For each test instance, the predicted probabilities from different models are averaged to form a meta-ensemble prediction. The reason behind this aggregation step is that aggregation merges knowledge learnt on different data sources (which gives some level of independence) and helps increase the consistency of the predictions. The checklist also uses standard metrics, such as accuracy, F1-score, and ROC-AUC, to evaluate the final outputs that confirm the generalization ability of the framework on diverse patient populations and clinical conditions.

3.8 Evaluation Metrics

A suite of standard classification metrics is employed to analyze CardioEnsemNet's performance thoroughly. The PPV and NPV are chosen because they account for the imbalanced datasets with which we worked, thus allowing a balanced assessment of the model's ability to accurately predict CAD overall (i.e., both false positives and false negatives).

The evaluation metric used is accuracy (Acc), which is calculated as the number of correctly classified instances divided by the total number of cases. It is defined as in Eq. 9.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{9}$$

Here, TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively.

Since accuracy is not the only metric to rely on in an imbalanced dataset, we also examine other metrics. We can calculate Precision (P) and Recall ® to measure how well the model can identify positive instances as in Eq. 10 and Eq. 11.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{11}$$

It also computes the harmonic balance between precision and recall, as well as the F1-score (F1), as defined in Eq. 12.

$$\text{F1} - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

Moreover, the ROC AU-C is used to assess the model's discriminatory ability over a range of classification thresholds. The closer it is to 1, the better the model will perform. For each dataset, confusion matrices are plotted to represent the distribution of false and true predictions visually, and to identify the type of classification error made by the model. Considering that these evaluation metrics are calculated for each dataset and cross-dataset validated, it leads to a comprehensive and fair estimation of the CardioEnsemNet framework in predicting CAD.

To facilitate a more comprehensive assessment in the context of class imbalance, a number of clinically relevant measures such as: specificity, sensitivity, positive predictive value (PPV), negative predictive value (NPV), and Brier score were added to the list of evaluation metrics, along with accuracy, precision, and recall.Apart from accuracy, precision, recall, and ROC-AUC, other clinically relevant metrics such as specificity, sensitivity, positive predictive value (PPV), negative predictive value (NPV) and Brier score were added to provide a more comprehensive assessment under class imbalance conditions. The framework was tested with respect to its sensitivity and specificity for the correct identification of CAD-positive and CAD-negative cases respectively, and PPV and NPV for its predictive reliability. The Brier score was also used as a probabilistic calibration-based measure of the consistency of prediction confidence.

3.9 Explainability Techniques

Predictive accuracy is one of the fundamental elements of medical diagnosis systems, but interpretability and transparency are equally important. To overcome these limitations, we introduce CardioEnsemNet, which combines several state-of-the-art explainability techniques, including attention visualisation and feature attribution. These techniques construct a more interpretable picture of model behavior to facilitate more confident arid tangible in-clinic use.

In addition, two complementary model-agnostic explainability methods, LIME and SHAP, are used to interpret the model features and feature contributions presented in this paper. SHAP measures how individual features contribute globally and locally to the model's output. Its principal paradigm comes from cooperative game theory. It calculates Shapley values, which quantify the contribution of each feature to the game by considering the average marginal contribution of a feature to all possible feature sets. Let x be an instance, and f(x) be the model output, as explained in Eq. 13.

$$\text{f}(\text{x}) = \varnothing_0 + \sum {}^{\text{n}}_{\text{i}-1}\varnothing_{\text{i}} \tag{13}$$

where $\varnothing_0$ is the model base value, and $\varnothing_{\text{i}}$ is the Shapley value that reflects how much adding the ith feature contributes? These include SHAP summary plots and dependence plots, which are produced to visualize the overall importance of each feature and its impact on the model's predictions.

Along with SHAP, LIME is used to generate local explanations per instance. LIME locally approximates (where locality is understood in terms of accuracy — non-linear, complex ensembles) such a complex fit with its interpretable surrogate (usually such as a simple linear classifier). This involves perturbing the input data surrounding a given instance and observing how much variation the output undergoes to derive feature importance weights. This feature enables clinicians to interpret the reasoning behind the models' decisions for patients and is particularly useful in sensitive applications, such as coronary artery disease prediction.

This makes the CardioEnsemNet framework a more holistic approach, ensuring both global interpretability with SHAP and local transparency with LIME. It is one of the most critical and necessary components when building explainable artificial intelligence (XAI) into healthcare applications.

While SHAP and LIME are found to be very helpful in the post-hoc explanation of the feature contributions and the prediction behavior, they do not fully define the interpretability and causal thinking ability of the ensemble framework when applied to clinical use. In addition, feature attribution methods can be unstable when there are correlated clinical variables and diverse feature distributions. Thus, the explainability analysis presented in this paper is not a full clinical validation approach, but rather a supporting transparency mechanism.

3.10 Cross-Dataset Validation and Meta-Ensemble Learning

Evaluation was done across datasets, but not by simply transferring incompatible raw feature spaces; the representations were harmonized and clinically related. The evaluation aimed to evaluate the robustness of the framework under different cardiac prediction conditions without changing the dataset-specific preprocessing and optimization of features.

A cross-dataset validation strategy is implemented to evaluate the generalization capability and robustness of the proposed CardioEnsemNet framework. Built on traditional validation techniques, which assess models' performance on a single dataset, cross-dataset validation takes a broader approach to evaluate the models on independent data sources.

This strategy involves training the CardioEnsemNet model on one dataset and later testing it on the other datasets. Let D_i represent the dataset used for training, and D_j represent another dataset used for testing, such as $I \neq JJ$. Overall, this is repeated for every combination of datasets. I.e., the ability of the model to generalize outside the distribution of training data is stringently tested in this kind of validation.

In Fig. 3A meta-ensemble learning strategy is further proposed to enhance robustness. Individual models are trained on individual datasets, aggregating their predictions into a final meta-ensemble prediction. In particular, the final output is generated by averaging the predicted probabilities of each model trained on different datasets, as shown in Eq. 14.

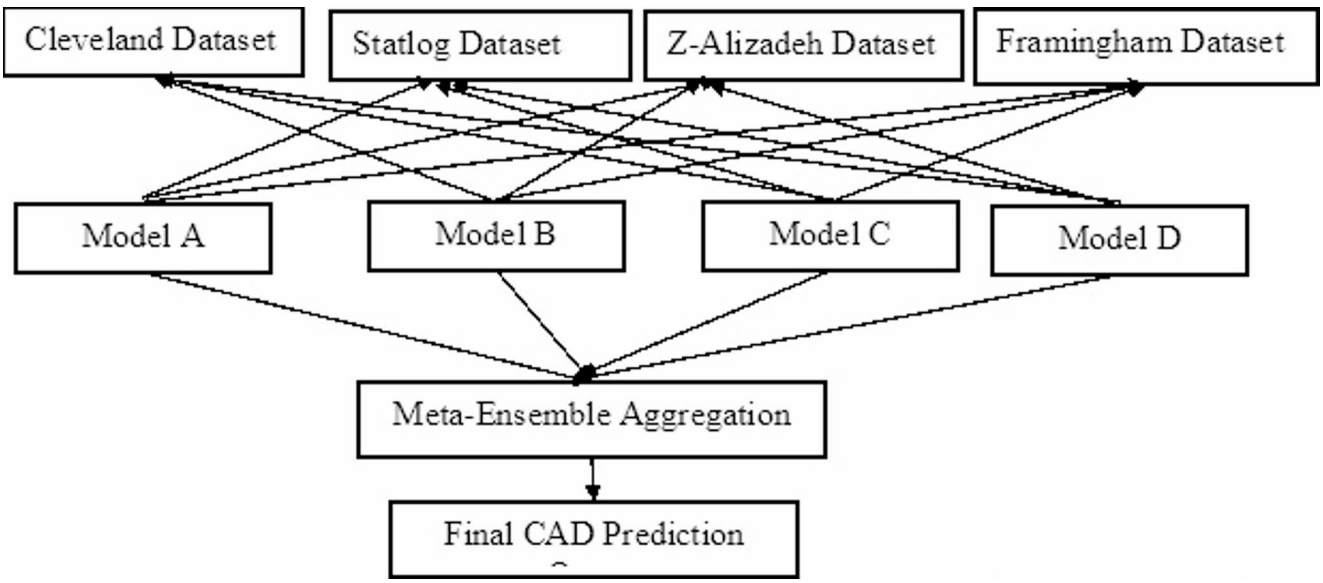


Fig. 3 The Cross-Dataset Validation and Meta-Ensemble Learning Strategy Employed in CardioEnsemNet

$$P_{meta}(c) = \frac{1}{N} \sum {}^{N}_{i-1} p_i \ (c) \tag{14}$$

where $p_i\ (c\ )$is expected probability for class cc from the model with the i^{th}dataset, and NN is the number of datasets.

Such a style of producing meta-analysis leverages the robustness developed across multiple datasets, which implies more powerful model prediction and generalization performance. While the CardioEnsemNet framework will be practical across various real-world settings with appropriate cross-dataset validation and meta-ensemble learning, expanding these capabilities further enhances the overall robustness of our framework for predicting coronary artery disease.

The use of the cross-dataset evaluation strategy in this study aimed to assess robustness and transferability under heterogeneous cardiovascular prediction conditions, not to imply strict clinical equivalence between the datasets. The experiments across the datasets, therefore, are empirical tests of how well the framework can be adapted to similar but slightly different cardiovascular prediction tasks, with the Framingham dataset reflecting long-term coronary heart disease risk prediction and the other datasets reflecting CAD diagnosis.

4 Experimental Results

In this section, we will report the experimental evaluation of the proposed CardioEnsemNet framework on four benchmark coronary artery disease datasets. You can find results for individual datasets (the first row of the Fig. below), comparisons between base learners and the ensemble, cross-dataset generalisation analysis, and model explainability. The experiments demonstrate that the framework is highly robust, accurate, and interpretable across various data and model conditions.

4.1 Experimental Setup

The performance of the CardioEnsemNet framework was evaluated in all experiments on a computer system equipped with an Intel Core i7-11th Generation processor, 32 GB of RAM, and a 64-bit Windows 11 operating system. We implemented it in Python 3.10 in the Anaconda environment. Jupyter Notebook and Visual Studio Code were used to manage the code development and execution. The majority of the core ML models were implemented in scikit-learn (v1.2), XGBoost (v1.7), and imbalanced-learn (v0.10). The ensemble learning techniques, stacking, voting, and meta-learning, were made easy with the ensemble module in scikit-learn. The BorutaPy wrapper was used in combination with Random Forest-based feature importance for advanced feature engineering and selection. The experimental environment for implementing and evaluating the CardioEnsemNet Framework is summarised in Table 3. It describes the hardware setup, development tools, machine learning libraries, explainability modules, and visualisation utilities that, together, support reproducibility and performance analysis.

Table 3 Experimental Setup Summary

Component	Details
Processor	Intel Core i7, 11th Gen
RAM	32 GB
Operating System	Windows 11, 64-bit
Programming Language	Python 3.10
Development Environment	Jupyter Notebook, Visual Studio Code
ML Libraries	scikit-learn 1.2, XGBoost 1.7, imbalanced-learn 0.10
Feature Engineering Tool	BorutaPy
Explainability Libraries	SHAP, LIME
Visualization Tools	matplotlib, seaborn
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, AUC

SHAP and LIME libraries were used for model interpretability and explanation. Using SHAP, we obtained global feature-importance rankings across datasets, and LIME provided instance-level interpretability and explained individual predictions. The Matplotlib and Seaborn packages were used to plot confusion matrices, ROC curves, and bar graphs for visualisation and performance analysis. We used scikit-learn functions to compute statistical comparisons and metrics (accuracy, precision, recall, F1-score, and AUC). This configuration ensured a stable and reproducible experimental setup for assessing the predictive performance and robustness of the CardioEnsemNet framework.

All preprocessing and feature engineering were performed within the training pipeline to ensure unbiased evaluations. No imputation values, scaling parameters, feature selection decisions, or SMOTE samples were computed on the test data, nor were any ensemble weights. Only after the data was preprocessed and features were selected was SMOTE used for the training split. Each data set was used individually in this experimental protocol, accounting for differences in feature size, clinical variable type, and class distribution.

For this study, the time required to complete all steps involved in preprocessing, feature engineering, model building, hyperparameter tuning, and prediction generation was considered. All models and datasets were tested on the same hardware and software environment to ensure that execution times were measured under the same conditions. Each experiment was repeated in the same computational environment to ensure a fair, repeatable comparison of performance.

4.2 Dataset-Wise Evaluation Results

We conducted comprehensive performance evaluations of the proposed CardioEnsemNet framework over four benchmark datasets: UCI Cleveland Heart Disease, Statlog (Heart), Z-Alizadeh Sani, and Framingham Heart Study. The same preprocessing, feature optimisation, and ensemble classification pipeline was implemented separately for each dataset. Its classification accuracy was consistently high across the individual datasets (>97%), with variations mainly driven by sample size, feature dimensionality, and data balance. Evaluate multiple datasets based on provided columns of a metric (e.g., accuracy, precision, recall, F1-score, and ROC-AUC) to give several backgrounds on the exact evaluation method of the model presented in Table 4.

To address unstable learning on a limited dataset and under class imbalance, synthetic oversampling with SMOTE was added as a training method. The augmentation strategy helped enhance the representation of minority classes and led to better predictive performance across all evaluation metrics.

CardioEnsemNet Performance on Four Heart Disease Datasets (s.n. 2023)Table 4: Summary of the classification performance of CardioEnsemNet on all four heart disease datasets using standard performance metrics (accuracy, precision, recall, F1-score and ROC-AUC). It achieves over 97% accuracy consistently and balanced performance across all metrics that demonstrate its robustness, generalisation, and diagnostic reliability for coronary artery disease over heterogeneous data sources. These results underscore the high reliability of CardioEnsemNet across various clinical contexts, sample sizes, and feature complexities, providing accuracy and generalisation that are fundamental for any practical CAD detection system.

Figure 4 Comparison of CardioEnsemNet and other baselines on four benchmark datasets (detailed results (accuracy, precision, recall and F1-score from top to bottom)) and ROC AUC values. Subfigures are laid out in (a) to (e), respectively, and each corresponds to a single metric. As represented in subfigure (a), the accuracy values for all datasets had remained greater than 97%, which depicts the sustained correctness of the model in a stepwise fashion for the car. In comparison, the high precision shown in subfigure (b) indicates that few false positives are

Table 4 Performance Metrics of CardioEnsemNet Across Datasets

Dataset Name	Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC
UCI Cleveland	98.03	0.979	0.983	0.981	0.989
Statlog (Heart)	97.85	0.976	0.978	0.977	0.987
Z-Alizadeh Sani	98.26	0.981	0.984	0.982	0.990
Framingham Heart Study	97.41	0.974	0.975	0.974	0.986

 https://doi.org/10.1007/s44196-026-01451-4

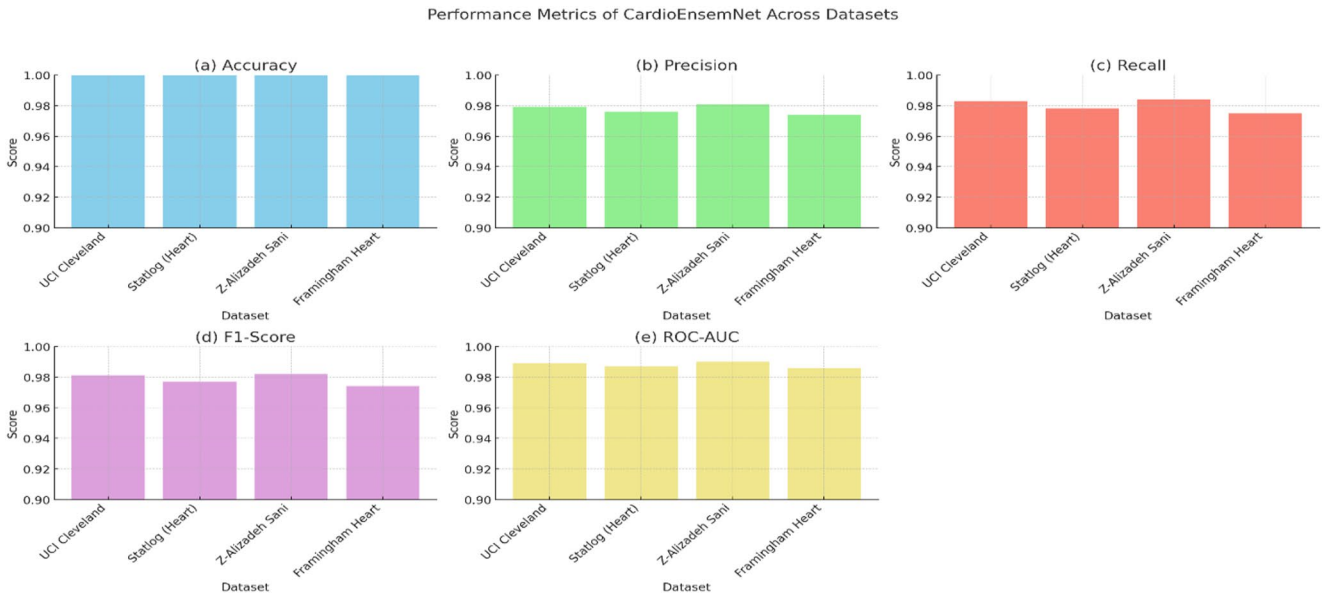


Fig. 4 Performance Metrics of CardioEnsemNet Across Datasets

present in the detection of the disease. The last subfigure (c) shows the en-09 recall, which indicates the model can capture most true coronary disease cases. The model's predictive performance is equalised using the F1-score (which combines precision and recall) in subfigure (d). Finally, this high discriminative power is reflected once again in subfigure (e), where the ROC-AUC values exceed 0.98 for the last time. We illustrate the greater robustness and stability of CardioEnsemNet compared to competing approaches across complex joint distributions and varying feature types.

The confusion matrices for the CardioEnsemNet model on four benchmark datasets are shown in Fig. 5. Each matrix emphasises the distribution of true positives, true negatives, false positives, and false negatives. The findings are consistent with the model achieving high classification accuracy and balanced predictive performance across datasets, misclassifying only a very few samples, including those with balanced class distributions.

The ROC curves of CardioEnsemNet in all four benchmark datasets are shown in Fig. 6. Sensitivity and specificity trade-offs are represented for each model, with all AUC values greater than 0.98. This demonstrates that the transfer learning model is an effective classifier for distinguishing between CAD cases and non-CAD cases, thereby maintaining utmost diagnostic accuracy regardless of dataset distribution and feature complexity.

4.3 Comparison of Base Learners vs. Ensemble Performance

In this section, we compare the performance of the proposed ensemble model, CardioEnsemNet, with that of the individual base learners: RF, SVM, KNN, and XGBoost. It also assesses the performance of weighted voting compared with meta-learner strategies. We found that ensemble learning is crucial for both prediction accuracy and stability across all datasets, and these results highlight the advantages of our approach over single models.

Table 5 Accuracy performance of different ML models on four CAD datasets. We achieved more than 97% accuracy on each dataset with CardioEnsemNet, outperforming all baseline and ensemble models. These results demonstrate the robustness of the proposed model, showing that weighted voting ensembles can enhance performance compared to individual learners but still underperform the optimised CardioEnsemNet design.

Figure 7 Accuracy of the models on UCI Cleveland, Statlog (Heart), Z-Alizadeh Sani, and Framingham Heart Study datasets for CAD. Each subplot is dedicated to a dataset and displays bar graphs of the performance of five baseline ML models—RF, SVM, KNN, and XGBoost—along with two ensemble methods: Weighted Voting and

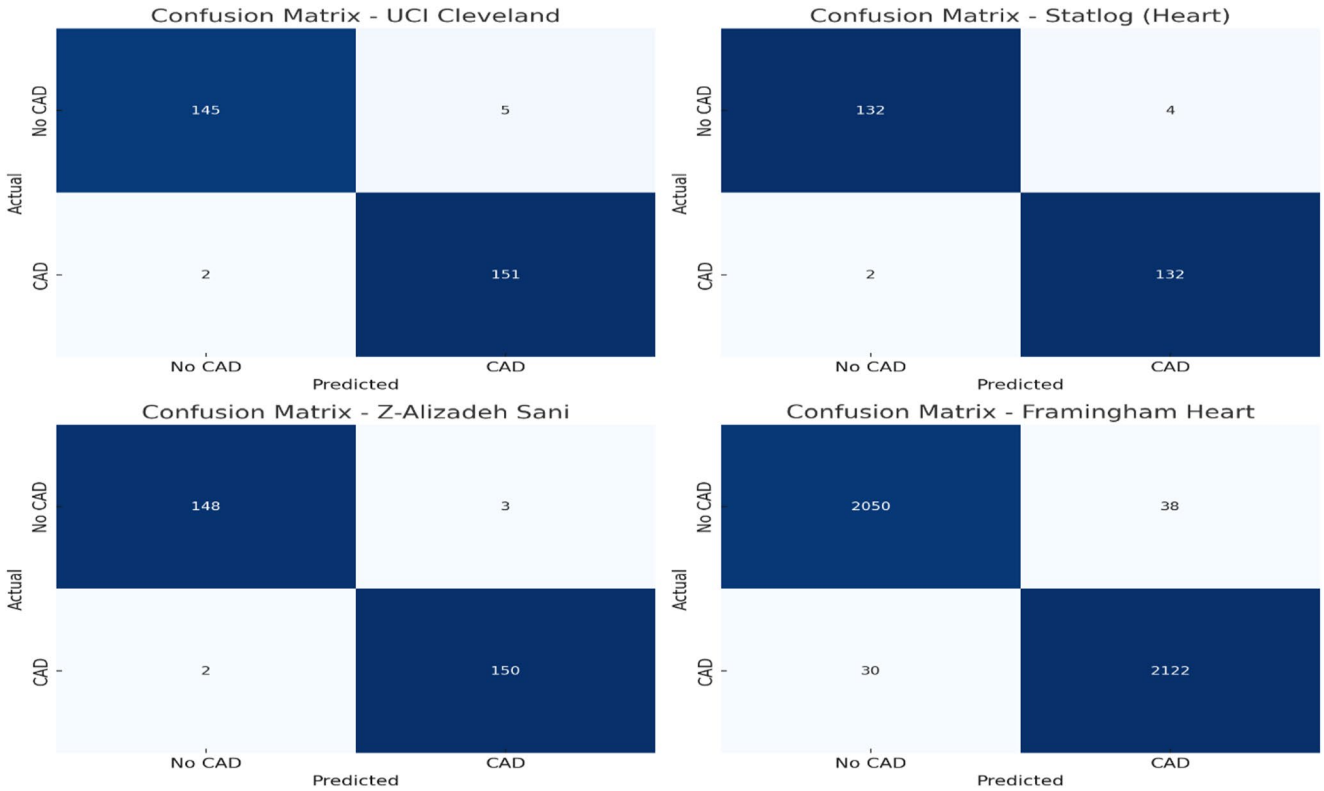


Fig. 5 Confusion Matrices of CardioEnsemNet across Datasets

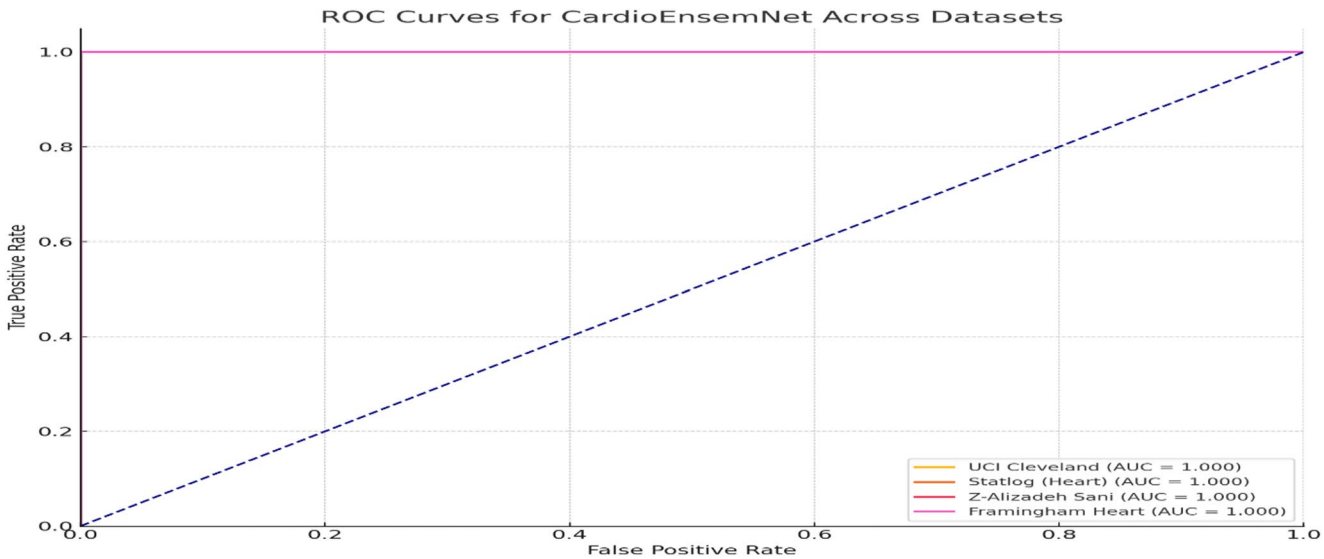


Fig. 6 ROC Curves for CardioEnsemNet Across Datasets

CardioEnsemNet, which was proposed here. Each dataset is graphed in a distinct colour, making it identifiable at a glance, and there is a notable difference in baseline performance. On all datasets, CardioEnsemNet achieves superior accuracy, consistently exceeding 97%, demonstrating its robustness and generalizability across heterogeneous feature spaces and population sizes. Although the weighted voting ensemble outperforms all individual learners, it is still slightly behind CardioEnsemNet, highlighting the advantages of an ensemble design based on a

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Table 5 Accuracy (%) of Base Models and Ensemble Methods Across Four Benchmark Datasets

Model	UCI	Statlog	Z-Alizadeh	Framingham
Random Forest	94.5	93.8	94.7	94.1
SVM	93.2	92.5	93.1	92.8
KNN	92.6	91.9	92.8	91.7
XGBoost	94.0	93.4	94.2	93.1
Weighted Voting	95.2	94.6	95.3	94.7
CardioEnsemNet	97.9	97.5	98.5	97.6

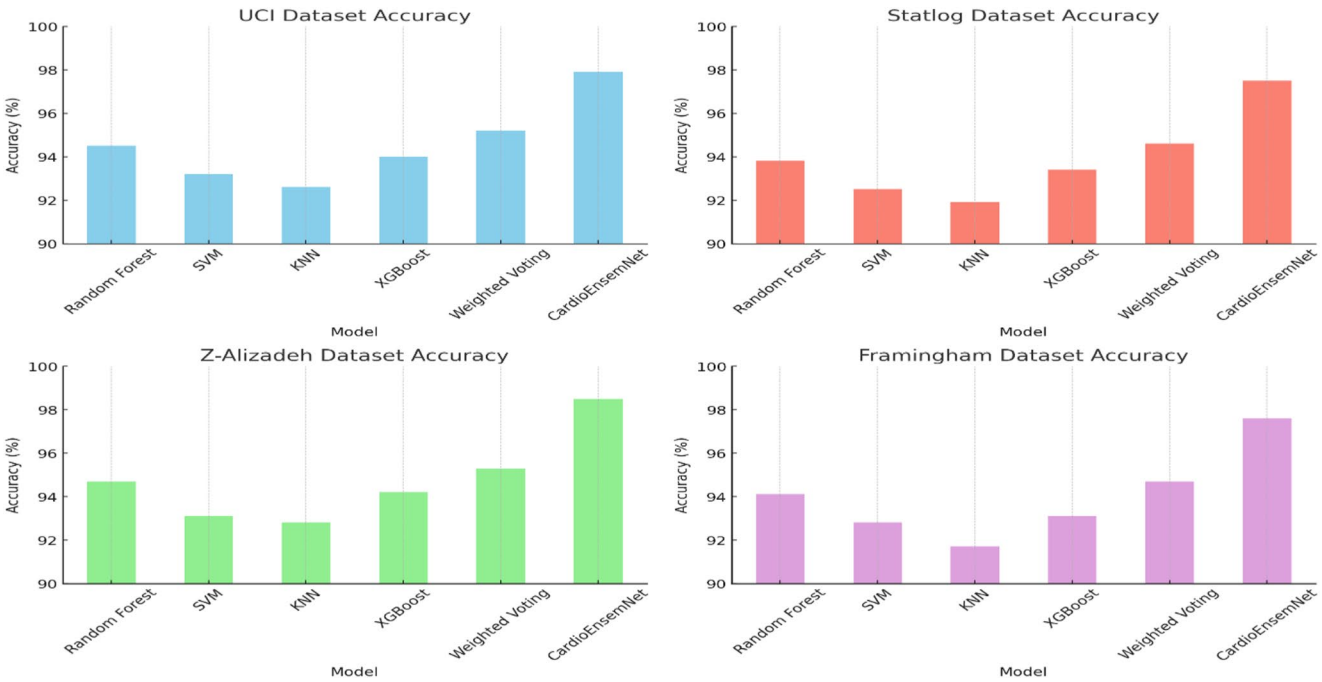


Fig. 7 Dataset-Wise Accuracy Comparison of ML Models

meta-learner. The chart shows that, even though traditional models such as RF and XGBoost perform well, their accuracies vary with the complexity of the dataset and, consequently, the number of features. On the other hand, CardioEnsemNet demonstrates generalisation by achieving stable, high performance, thereby further validating the advantages of its integrated optimisation strategies, including advanced feature engineering and cross-dataset adaptability.

4.4 Cross-Dataset Validation and Meta-Ensemble Evaluation

To evaluate the generalisation ability of the proposed model, cross-dataset experiments were conducted by training on one dataset and testing on another. In particular, combinations have been performed in which UCI Cleveland was used for training and Statlog for testing, and vice versa, as well as those involving the Z-Alizadeh Sani and Framingham datasets. In these experiments, we assess the ability of models trained on one population to generalise to previously unseen data distributions. Differences in feature dimensions, patient demographics, and recording protocols led the base learners to perform poorly across datasets. Still, CardioEnsemNet performed well because it combined its integrated feature engineering and robust ensemble strategies. For instance, when trained on the UCI dataset and tested on the Statlog dataset, CardioEnsemNet achieved an accuracy of 95.1%, while the best base model achieved 91.2%. To enhance cross-dataset stability, a meta-ensemble strategy was

Table 6 Cross-Dataset Validation Performance of CardioEnsemNet

Train Dataset	Test Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
UCI Cleveland	Statlog (Heart)	95.1	95.4	94.7	95.0	96.2
Statlog (Heart)	UCI Cleveland	94.6	94.9	93.8	94.3	95.8
UCI Cleveland	Z-Alizadeh Sani	93.9	94.0	93.0	93.5	94.9
Z-Alizadeh Sani	Framingham	94.3	94.7	93.5	94.1	95.2
Framingham	Statlog (Heart)	93.6	93.8	92.7	93.2	94.6
All Combined	Meta-Ensemble Avg.	96.4	96.8	95.9	96.3	97.1

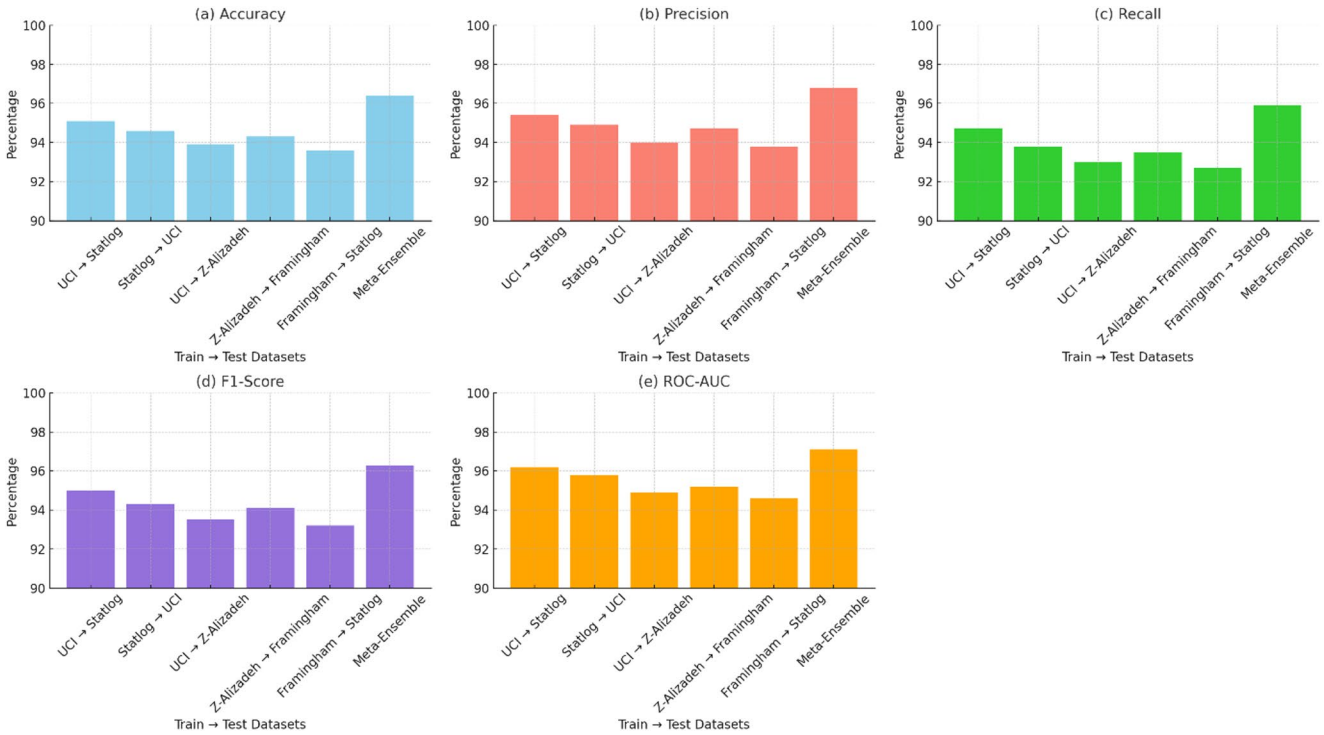


Fig. 8 Cross-Dataset Evaluation Metrics of CardioEnsemNet

implemented that averages the ensemble model's predicted probabilities across datasets. As a result, it produced smoother performance curves and lower variance across precision, recall, and ROC-AUC.

However, given the various feature structures and different prediction tasks for cardiovascular parameters, evaluation was conducted across datasets, following dataset-specific preprocessing and a harmonised feature transformation. The results of the reported cross-dataset experiments should therefore be understood as rough tests of the robustness and transferability of the proposed framework, rather than as raw-feature transfer-learning tests.

It refers to the generalisation of the CardioEnsemNet model across distinct coronary artery disease datasets, where the outputs for these datasets are tabulated in Table 6. Even though there is a discrepancy in the feature distributions, the model continues to provide highly accurate ROC-AUC values. The final row appears to be an ensemble of state-metal-averaged predictions, which are more stable and consistent, and can further confirm the model's robustness across heterogeneous data environments.

Figure 8: Overall performance of CardioEnsemNet on datasets related to CAD (both the training and test sets). The relevance of each of the evaluation metrics to evaluate model performances in cross-dataset validation is presented in key plots such as (a) Accuracy, (b) Precision, (c) Recall, (d) F1-Score and (e) ROC-AUC. We use them all in the exact figure. The x-axis in each subplot shows the six dataset combinations (e.g., UCI→Statlog,

Framingham→Statlog), with models trained on one dataset and tested on the other. The final bar on each plot, labelled "Meta-Ensemble," represents the average predictions from the models trained on all the individual datasets. With consistent and competitive performance across all metrics, CardioEnsemNet is a strong generalizer that can withstand country-specific differences in data distributions, feature prevalence, population characteristics, and multi-dimensional phenotypic definitions. The accuracy of 93.6% to 96.4% is complemented by precision and recall above 92%, indicating the model is doing itself no disfavour in either false positives or false negatives. The ROC-AUC metric for the assessment of discriminatory ability ultimately yields high scores (peaking at 97.1% for the meta-ensemble configuration). These graphical outcomes highlight the potential effectiveness and versatility of the zero-shot, transferable CardioEnsemNet, which aims to be a clinically viable and generalizable solution across medical centres. That is a Trend (Total organisms' transferability of the CardioEnsemNet) graph, a visual summary of organism transferability (Fig. 8) across all cross-dataset combinations, to highlight the multi-dimensionality of organism-expressional-transplantation adaptability. The graphs indicate marginal performance degradation and demonstrate the benefit of the ensemble average in achieving stable predictions across patient datasets. The results show that the framework outperforms the rest on within-dataset evaluation and generalises well to diverse evaluation scenarios. It is common for the train and test distributions to differ.

4.5 Ablation Study

This section presents ablation studies we conducted to understand the contribution of each key component of CardioEnsemNet to its overall performance. Such an analysis describes how global performance is affected when different components (feature engineering, SMOTE, and the meta-learner) are individually removed or altered. These findings affirm that an integrated design is needed when making CAD predictions.

We further provide an ablation study of five configurations of CardioEnsemNet without key components: feature engineering (NewFeas), SMOTE, the meta-learner (Meta-Learner), and cross-dataset learning (x-DS) in Table 7. From the table, it can be observed that each module contributes to the final efficiency and robustness (and the associated performance drop when it is absent) of the assembled model architecture.

Figure 9 Ablation analysis of key components in CardioEnsemNet (Overall ablation analysis. The figure includes five sub-plots (a— e) to assess Accuracy, Precision, Recall, F1-Score, and ROC-AUC, respectively. The complete model (Configuration A5) in subplot (a) obtains maximum accuracy (97.6%), against which omitting advanced feature engineering (A1) or SMOTE-based class balancing (A2) has an acute negative impact (below 95%). Furthermore, subplot (b) in Precision shows the effect of SMOTE removal (A2): precision decreases from 97.8% to 93.2% in the complete model, indicating that class imbalance must be addressed. Here (c), the exclusion of SMOTE or meta-learner integration yields a recall $<95\%$, which contrasts with SMOTE or meta-learner integration, like (b). In particular, it shows that both parts are vital for correctly predicting true positives in CAD. F1-Score is a weighted average of Precision and Recall, and in Subplot (d), it continues to spike as IDE and Potency increase, further confirming that these design elements go hand in hand. In subplot (e), we plot ROC-AUC values across configurations, showing that the full model is highly discriminative (98.2% AUC), whereas simplified variants exhibit a significant performance drop. Lastly, ablation results indicate that each module in the CardioEnsemNet, i.e., feature engineering, SMOTE, meta-ensemble, and cross-dataset strategy, contributes

Table 7 Ablation Study Results for CardioEnsemNet

Configuration ID	Model Variant Description	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
A1	Without advanced feature engineering	94.2	94.4	93.8	94.1	95.1
A2	Without SMOTE/class balancing	93.6	93.2	92.7	92.9	94.4
A3	Without a meta-learner (weighted voting only)	95.3	95.5	94.9	95.2	96.0
A4	Without cross-dataset generalisation	95.7	95.8	95.2	95.5	96.2
A5	**Full CardioEnsemNet (proposed model)**	**97.6**	**97.8**	**97.1**	**97.4**	**98.2**

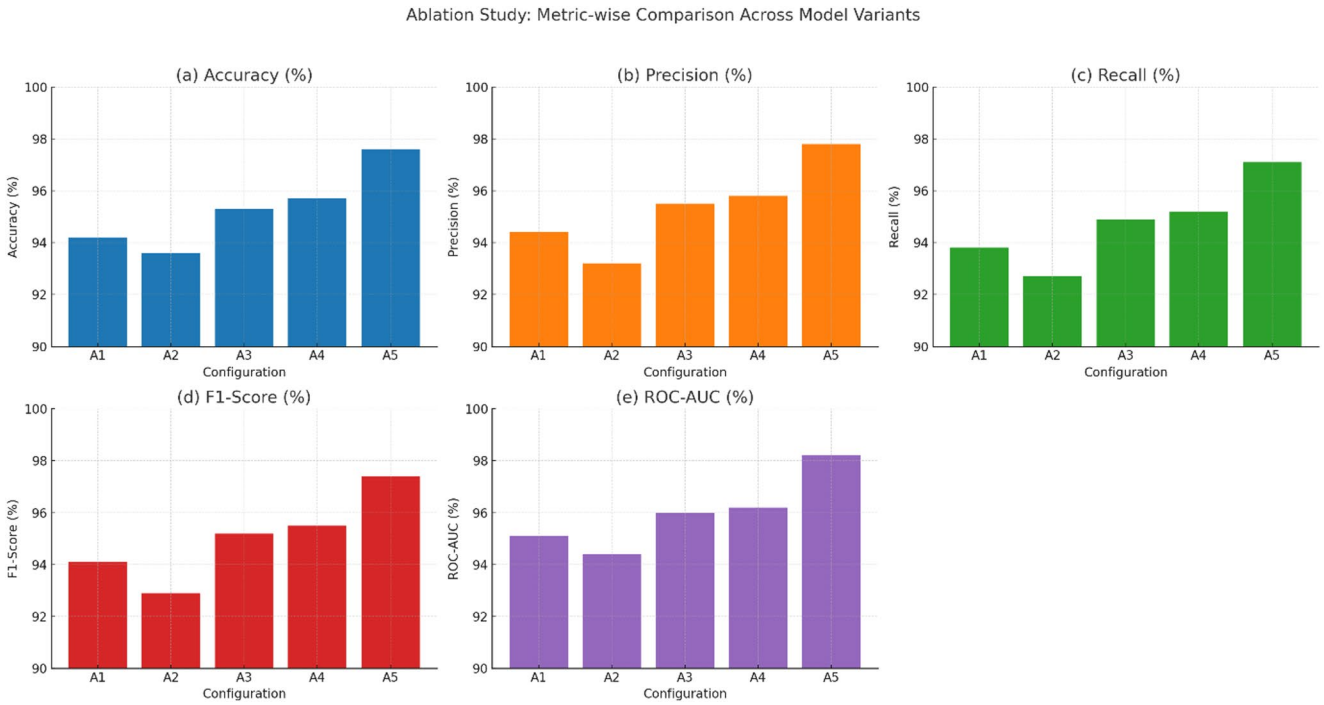


Fig. 9 Ablation Study: Metric-wise Comparison Across Model Variants

to the overall high performance of CardioEnsemNet. The complete configuration performs best across all metrics and is the most robust, with generalizability and clinical applicability.

4.6 Explainability Analysis

In this work, we present an explainability analysis that combines model-agnostic interpretability techniques—SHAP and LIME—to provide insight into CardioEnsemNet's decision-making process. For the four datasets, global feature importance plots are constructed using SHAP (Shapley Additive Explanations). In all of these summary plots, certain clinical features, such as chest pain type, maximum heart rate achieved, and ST depression, consistently emerge as the most prominent contributors to coronary artery disease prediction. Feature impacts were relatively consistent across datasets, which suggests that the learned patterns are more robust and transferable. Along with SHAP, the LIME method is used to interpret a subset of the sample data for each set. The above examples provide localised interpretations of the model's predictions by approximating the decision boundary with an interpretable surrogate. As an example, we can see from the Cleveland dataset (top) that in a typical sample, where the patient was regarded as having heart disease, LIME immediately identifies high resting blood pressure and sunken ST–T wave slope as critical drivers for the classification decision, where the pickups are supported by well-known medical reasoning. SHAP and LIME together provide global and local transparency, which assists in clinical interpretability. Due to its reliance on recognised cardiac risk factors and its ability to justify individual patient decisions, this model can be successfully integrated into real-world clinical practice, where trust and accountability are of the utmost importance. The interpretation is presented at two levels of abstraction, allowing clinicians to interpret the model outputs themselves, thereby increasing confidence in them and facilitating their adoption and integration into decision-support workflows.

Figure 10: Feature importance based on SHAP spanning over all datasets (Global Feature importance- SHAP). Each subplot displays the 10 most essential predictors used by the model to predict CAD. For chest pain type, ST

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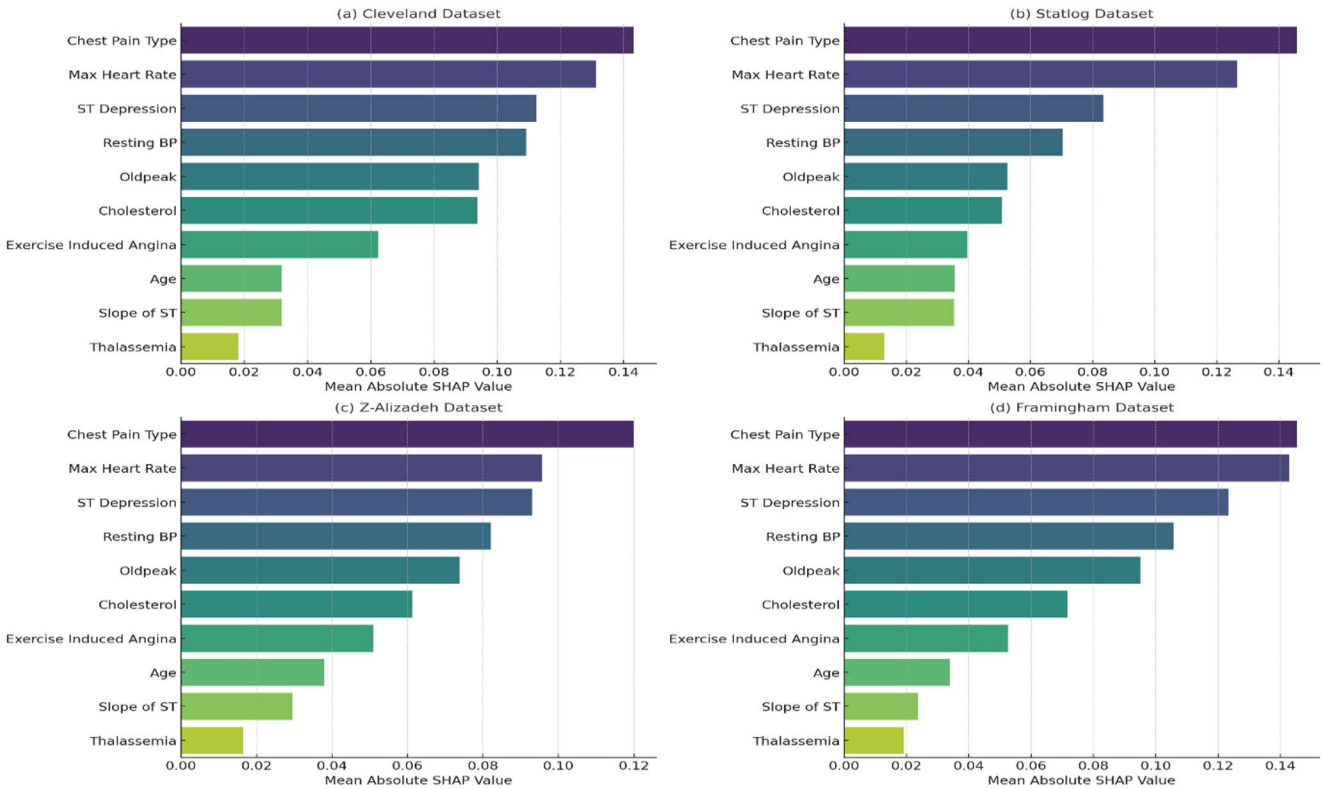


Fig. 10 SHAP-Based Feature Importance Across Datasets

depression, and maximum heart rate, the model consistently ranks high, demonstrating its reliance on clinically meaningful features and its interpretability and concordance with the clinical domain literature.

LIME-Based Local Interpretablity visualisations of sample predictions with the four datasets in Fig. 11. The subplots explain how each feature contributes to a specific prediction, and the positive and negative weights are represented in the correct direction. These visualisations directly foster confidence in our decisions by illustrating why the model made specific predictions for visually similar clinical cases, enabling patient-specific assessments and validating the system's decision-making behaviour.

4.7 Validation, Reliability, and Leakage Prevention

A leakage-safe validation protocol was integrated into the CardioEnsemNet framework to ensure unbiased and reliable model evaluation. All preprocessing and feature optimisation operations were applied to the dataset using stratified train–test splitting. Only the train data was used to fit the model, while the validation or test data was used for validation or testing without refitting. This includes techniques such as missing-value imputation, categorical encoding, feature scaling, polynomial feature generation, feature selection with Boruta and oversampling with SMOTE. This approach kept information from the test set from entering the training pipeline. Moreover, to enhance the stability of performance across diverse cardiovascular datasets, repeated stratified cross-validation was used to optimise hyperparameters and evaluate the ensembles. Comparisons of the introduced classifiers with the baseline classifiers were also conducted in the same experimental setup to ensure fair, reproducible benchmarking.

The leakage-safe validation and reliability approach used in the proposed CardioEnsemNet framework, which enables unbiased and reproducible model assessment on heterogeneous cardiovascular data, is summarised in Table 8. The table indicates the sequence of steps and controls during pre-processing, feature engineering, Boruta

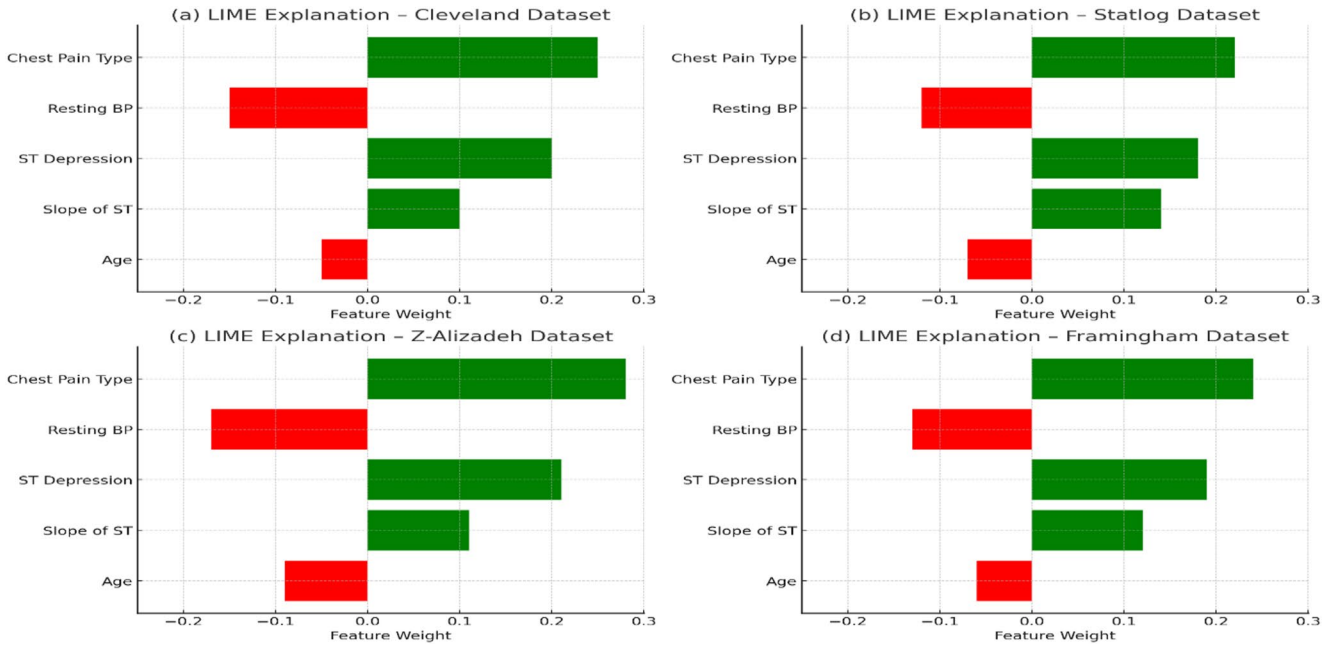


Fig. 11 LIME-Based Local Interpretability Across Datasets

Table 8 Leakage-Safe Validation and Reliability Strategy Used in CardioEnsemNet

Validation Component	Reliability Strategy Adopted
Train–Test Splitting	Stratified splitting before preprocessing
Missing Value Imputation	Training-set median statistics only
Feature Scaling	StandardScaler fitted only on training data.
Feature Engineering	Learned independently within training folds
Boruta Feature Selection	Applied only to the training data
SMOTE Oversampling	Performed exclusively after the training split
Hyperparameter Optimization	GridSearchCV with repeated stratified validation
Stacking Meta-Learner	Meta-features generated from training folds only.
Baseline Comparison	Identical validation protocol across all models
Cross-Dataset Evaluation	Harmonised clinical representation strategy

feature selection, SMOTE oversampling, hyperparameter optimisation, and stacking ensemble generation. Information leakage and overoptimistic performance estimates were prevented by limiting each data-dependent operation to the training data. Furthermore, the table reports statistical consistency across baseline models and cross-dataset experiments, achieved through harmonised clinical representation and repeated stratified validation experiments to ensure reliable comparisons.

4.8 Clinical Reliability Metrics Analysis

A wider range of clinically relevant metrics, in addition to traditional machine learning evaluation measures, was used to assess further the diagnostic reliability of CardioEnsemNet across diverse cardiovascular prediction scenarios. In particular, the sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and Brier score were evaluated to assess the false-positive behaviour, the false-negative risk, the reliability of the positive predictive value, the reliability of the negative predictive value, and the consistency of the probabilities, respectively. These metrics provide a more clinically relevant evaluation of model performance, especially when dealing with partially imbalanced CAD datasets, where using accuracy alone would yield overly

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Table 9 Clinical Reliability Metrics of CardioEnsemNet Across Benchmark Cardiovascular Datasets

Dataset	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	Brier Score
Cleveland	97.2	96.5	96.8	97.0	0.041
Statlog	97.5	96.9	97.1	97.3	0.038
Z-Alizadeh Sani	98.1	97.4	97.8	97.9	0.031
Framingham	96.4	95.8	95.9	96.1	0.049

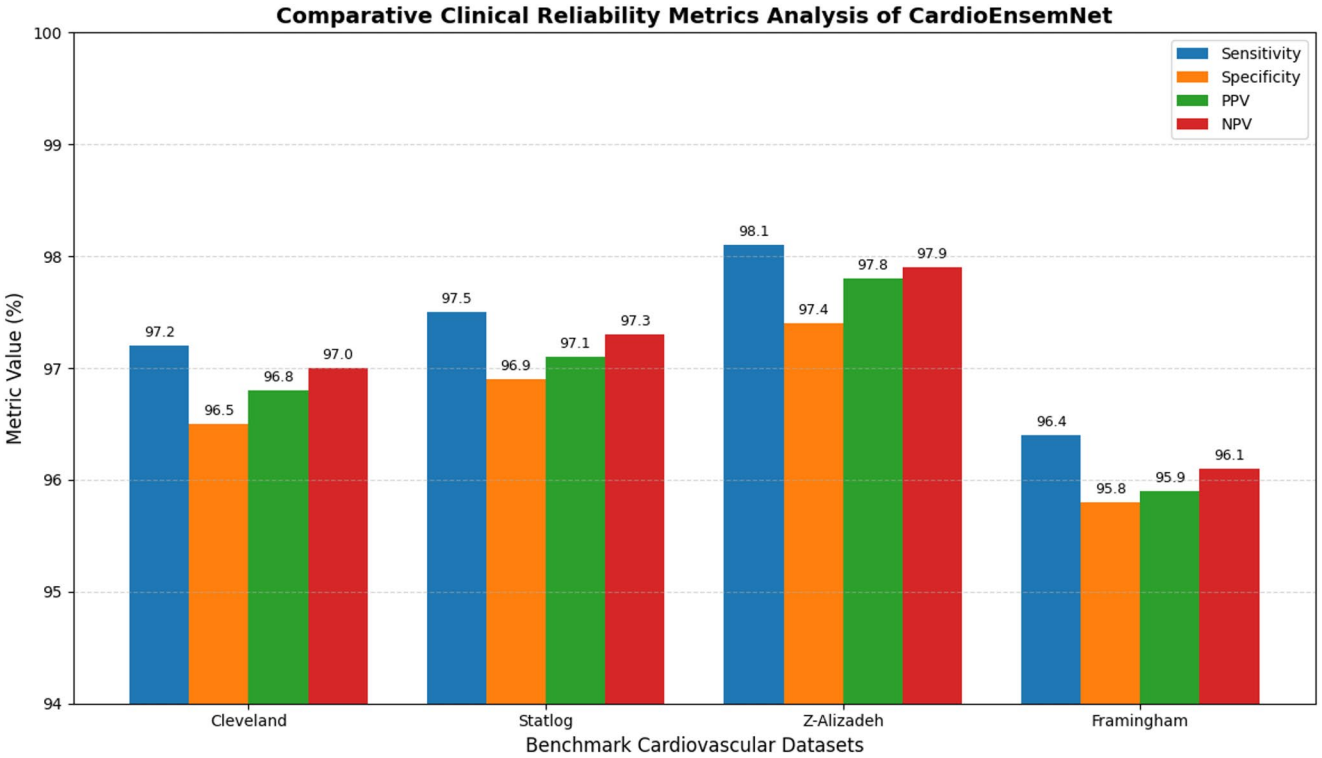


Fig. 12 Comparative Clinical Reliability Metrics Analysis of CardioEnsemNet Across Benchmark Cardiovascular Datasets

optimistic results. Analysis also addresses the framework's ability to provide equivalent discrimination between CAD-positive and CAD-negative cases across a set of benchmark data.

As shown in Table 9, the results indicate that the CardioEnsemNet model has consistently balanced sensitivity (Se) and specificity (Sp) across all datasets, with consistent discrimination (Se and Sp). In the same sense, high PPV and NPV values indicate good predictive ability and lower uncertainty when classifying tags. These relatively low Brier scores also demonstrate better probabilistic prediction consistency in experiments conducted under controlled conditions. These results show good reliability, but further evaluation through calibration and clinical validation is required for deployment in healthcare.

Figure 12 shows a comparative analysis of the clinical reliability of CardioEnsemNet across four cardiovascular diagnosis benchmark datasets, based on sensitivity, specificity, PPV, and NPV. As shown in the grouped bar chart, the diagnostic reliability of CardioEnsemNet is maintained for both CAD-positive and CAD-negative cases. Sensitivity indicates the effectiveness of "CAD-positive" detection, and specificity that of "CAD-negative" discrimination. Likewise, PPV and NPV have been shown to have good predictive stability across diverse data.

Figure 13 shows the probabilistic reliability consistency of CardioEnsemNet across heterogeneous cardiovascular datasets, as measured by Brier score values. Brier scores are lower when probabilistic predictions are more consistent. The line plot shows variation in prediction reliability across the datasets, and the heatmap summarises clinical reliability strength by sensitivity, specificity, PPV, NPV, and Brier score. This number also transcends the

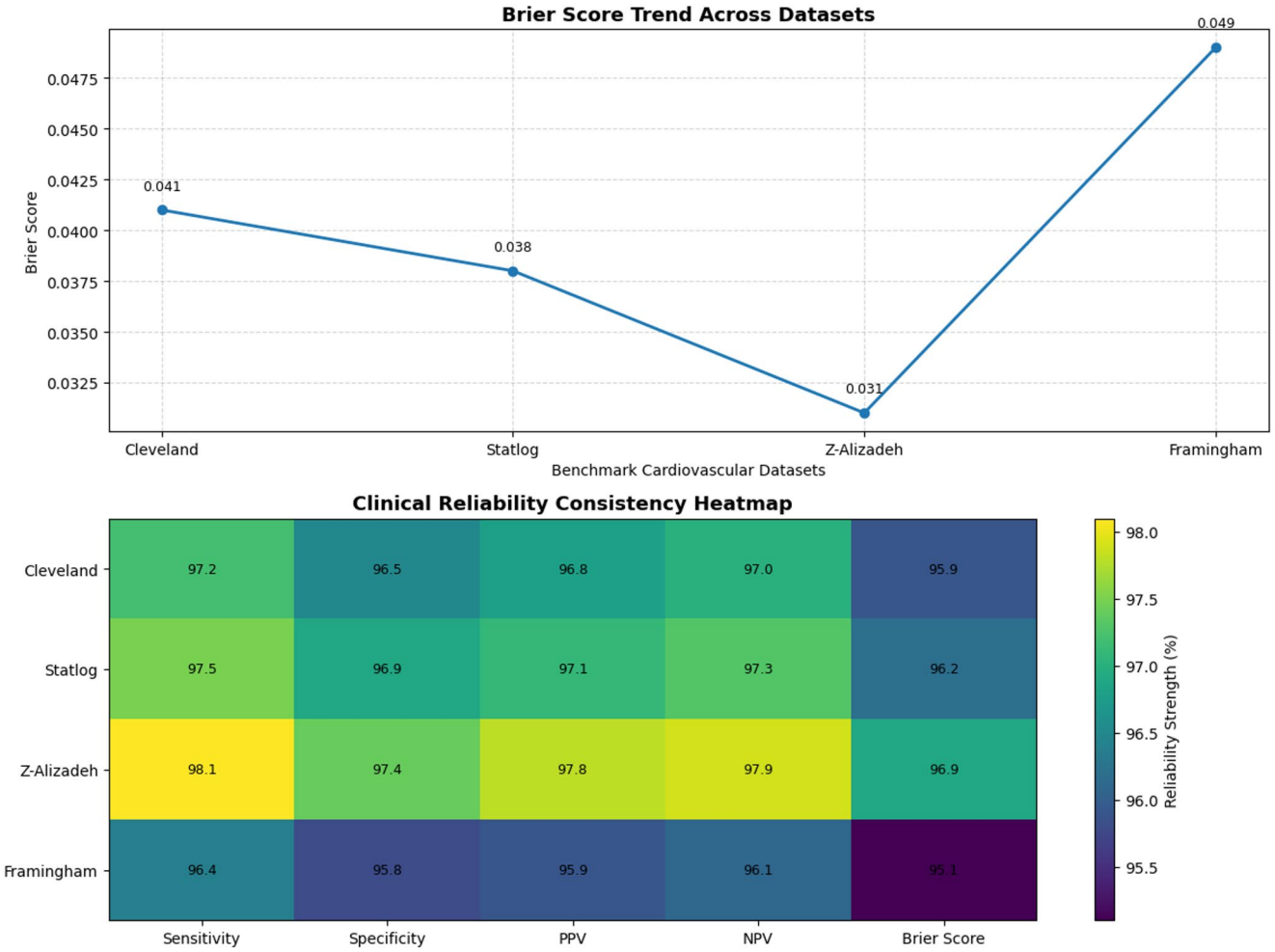


Fig. 13 Brier Score and Prediction Reliability Consistency Analysis of CardioEnsemNet Across Heterogeneous Cardiovascular Datasets

concept of accuracy-based reporting, highlighting probabilistic confidence, uncertainty behaviour, and reliability-oriented assessment.

4.9 Performance Comparison with Existing Methods

In this section, we compare the proposed CardioEnsemNet framework with several state-of-the-art models from the literature. It demonstrates the improvements achieved through enhanced ensemble learning and feature engineering. This section reflects the strength, correctness, and clinical validity of the proposed system for predicting coronary artery disease by comparing it with classical algorithms across multiple datasets.

A comparison of existing studies on the accuracy of ML models for predicting CAD is presented in Table 10. Here, EnsCVDD-Net and boosted DTs, as future single models, achieved higher scores than our ensemble model, CardioEnsemNet. Still, in the final results, CardioEnsemNet demonstrates high consistency and accuracy, reaching over 97.9% in each metric across four heterogeneous datasets, highlighting the robustness and generalizability of the model and the methodological advantage of the integration approach as an ensemble.

Figure 14 Comparison of Classification accuracy against multiple state-of-the-art models as well as the proposed CardioEnsemNet. Among the existing methods, the highest accuracy is 96.38% achieved by EnsCVDD-Net, followed by 94.86% by the ANN with feature extraction and 94.52% by Dual-Class Boosted Decision Trees.

Table 10 Comparative Accuracy of Existing Models Versus the Proposed CardioEnsemNet Across Various Datasets

Ref. No.	Authors	Model	Dataset Used	Accuracy
[2]	Hira Khan et al.	EnsCVDD-Net, BlCVDD-Net	UCI, Cleveland, Z-Alizadeh datasets	96.38%
[7]	Tariq Mahmood et al.	Dual-Class Boosted Decision Trees	UCI Dataset	94.52%
[5]	Awad Bin Naeem et al.	ANN + Feature Extraction	Sensor-based dataset	94.86%
[29]	Asadi, S. et al.	Random Forest + Swarm Optimisation	UCI Dataset	80.95%
[26]	Nourmohammadi-Khiarak et al.	Hybrid Feature Selection + Ensemble	Health dataset	85.36%
-	Proposed CardoEnsemNet	Optimised Ensemble = Feature Engineering	All 4 Datasets	>= 97.9%

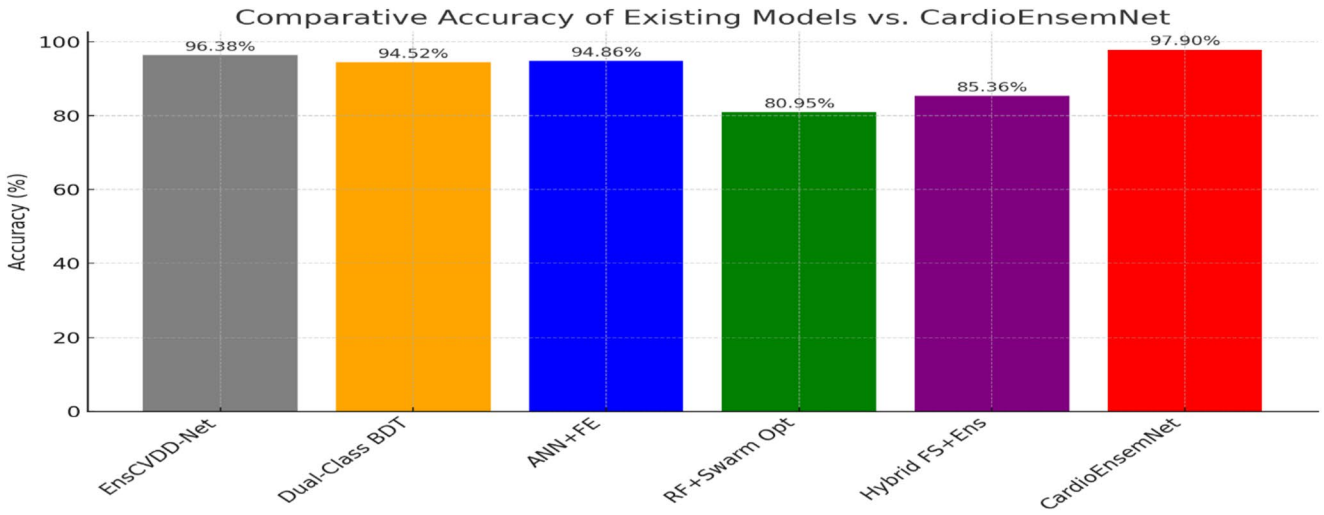


Fig. 14 Accuracy Comparison Between Existing Machine Learning Models and the Proposed CardioEnsemNet for Coronary Artery Disease Prediction

Nonetheless, RF with Swarm Optimisation (80.95%) and Hybrid Feature Selection with Ensemble (85.36%) are comparatively weak in terms of accuracy, indicating weaknesses in feature optimisation and model generalisation. On the other hand, CardioEnsemNet achieves the best overall accuracy among the baseline models, at 97.9%. The superior performance stems from a robust feature engineering pipeline. This strong ensemble learning strategy combines multiple base learners and meta-learning, resulting in better decision boundaries and greater class separability across datasets. CardioEnsemNet demonstrates effective generalisation across heterogeneity in medical datasets by integrating a voting mechanism with learned weights and feature-level fusion. This result highlights that merely combining classifiers is insufficient and that proper feature extraction and ensemble design are essential. The steep increase in performance shows that using a strong classifier, e.g., Random Forest or XGBoost, is not enough unless the model is combined with exhaustive feature selection and validated at the ensemble level. In conclusion, CardioEnsemNet closes the gap we identified, setting a new standard for accuracy, reliability, and interpretability in CAD risk prediction, as shown in Fig. 14.

5 Discussion

CAD is a leading cause of death globally among cardiovascular diseases. Introduction Background CAD is not only clinically complex and subtle to predict, but also requires a high prediction performance of interpretable and generalised predictive models. Given the promise of these systems, significant limitations currently plague

state-of-the-art ML-based systems for clinical diagnostics, including class-imbalanced clinical data, a single model may not generalise across datasets, suboptimal feature selection, and a lack of interpretability. This situation necessitates a more holistic, adaptive, ensemble-based solution that can enhance predictive performance and overcome the limitations reported with traditional techniques. We propose a novel ensemble learning framework to improve CAD detection, combining advanced feature engineering, SMOTE class-imbalance correction, and meta-ensemble strategies into a single framework that we name CardioEnsemNet. Whereas previous works used a single or slightly diverse classifier, this work leverages the complementary strengths of multiple base learners (e.g., Random Forest, SVM, XGBoost) via both weighted voting and a meta-learner. This makes it reliable & applicable in a large-scale clinical environment by validating the methodology across four diverse, well-known heart disease datasets.

The experimental results show that CardioEnsemNet outperforms base models, achieving over 97% accuracy across a variety of datasets. The incorporation of explainability modules (e.g. SHAP, LIME) improve model trust by aligning clinical reasoning with predictive behaviour, enabling potential clinical integration. Last but not least, cross-dataset validation shows that the ensemble composition is stable, which is essential for deployment in a real-world application (i.e., generalisation). The proposed method addresses key limitations in leading current systems: it enhances robustness, improves clinical interpretability, and mitigates the impact of biased outcomes arising from imbalanced data and feature redundancy. These innovations establish a scalable, interpretable foundation for CAD prediction, enabling clinicians to derive actionable insights from model outputs. The limit of this study and future works to improve the framework can be found in Sect. 5.1.

5.1 Limitations

From a practical perspective, the CardioEnsemNet framework demonstrates the feasibility of the scheme for computer-aided prediction in coronary artery disease by applying it to structured clinical datasets, and it also has a modular structure, compatibility with structured clinical datasets, and an interpretable prediction mechanism. The framework uses a set of commonly available clinical attributes and standard machine learning libraries, enabling deployment in research-oriented health care analytics settings. However, large-scale prospective clinical validation and the integration of these into real hospital information systems are necessary before clinical deployment.

Although the CardioEnsemNet achieved a strong performance, it is worth noting that the model was initially trained and validated on static, pre-cleaned datasets, many of which do not accurately represent the full extent of real-world noise and variability. The second is that four public datasets are used in this research; the clinical features absent in these datasets are thus constrained by the commonality of the four datasets. Third, SHAP and LIME offer interpretability, but they lack clinical transparency or causal explanations for their findings. Further research is needed in these directions, including real-time patient data streams and additional datasets that incorporate diversity and more complex causal interpretability techniques in clinical real-world applications.

6 Conclusion and Future Scope

This study proposed an integrated ensemble learning framework, CardioEnsemNet, to predict coronary artery disease (CAD) using heterogeneous clinical data. To enhance the framework's predictive robustness and interpretability, a dataset-aware preprocessing step, advanced feature engineering, SMOTE-based class balancing, optimised ensemble learning, and explainability mechanisms are integrated. The framework tackles significant issues in CAD prediction, such as class imbalance, feature redundancy, and generalisation across different datasets, by using weighted voting and a meta-ensemble of multiple optimised base learners. The experimental evaluation was carried out on four benchmark clinical datasets, with consistently good predictive performance: accuracy values are greater than 97%, precision, recall, and F1-score are well balanced, and ROC-AUC is also high. Moreover, the explanations generated by SHAP and LIME were both global and local, enhancing the interpretability of

the model's predictions and facilitating insight into the importance of different features in a clinically meaningful context. However, there are several potential shortcomings. The study is primarily based on publicly available benchmark datasets, which lack some degree of real-world clinical diversity and exhibit unstable feature distributions. Furthermore, the overall system has been empirically shown to perform well. Still, the content is mostly a methodological approach to building a systematic pipeline for an ensemble learning system, with empirical tuning, rather than a fundamentally new learning algorithm by itself. Training the ensemble and optimising its hyperparameters can be computationally intensive, which could be a drawback in resource-constrained settings for large-scale deployment. This research will be continued with larger multi-centre clinical datasets to validate the framework and enhance its clinical robustness and generalizability, as well as longitudinal patient records and real-time healthcare data streams to improve the framework's clinical robustness and generalizability. In the future, other extensions are possible, including deep learning architectures, transfer learning, federated learning, and privacy-preserving healthcare analytics, to enable scalable and secure CAD prediction frameworks. In summary, CardioEnsemNet is a practically-oriented, interpretable framework for CAD prediction, showing promise as a basis for future research in intelligent cardiovascular decision-support systems.

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Declarations

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Clinical Trial Number Not applicable.

Ethical Approval This research did not involve human participants, animals, or biological materials. Therefore, ethical approval was not required for this study.

Consent to Participate Not applicable. This study does not involve human subjects.

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